RAFT-Ports

Risk **A**ssessment **F**or **T**sunamis in **Ports**



User Manual

RAFT-Ports: Risk Assessment For Tsunamis in Ports

The app and user manual are created by Raquel P. Felix. Microsoft 365 Copilot was utilised to assist with coding tasks, improve efficiency, and organise the overall structure of the codebase

This document provides a general guide on how to install and use the applications, along with additional notes on preparing your input in QGIS and how the RAFT-Ports results are validated using QGIS and Excel.

The general layout of this manual and some of the wording used for the instruction buttons are based on the user manual created by Felix et al. (2025).

Please cite the following paper when using the application:
Felix R., Chua C.T., Suppasri A., Pranantyo I. R., Gunawan, E. and Heidarzadeh M. (2026). RAFT-Ports: An Application for Risk Assessment for Tsunamis in Ports Demonstrated at Tanjung Gerem Port, Indonesia. *Geosci. Lett. 13, 74 (2026)*.
https://doi.org/10.1186/s40562-026-00522-w

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Installation and Setup

The application is developed using the App Designer within the MATLAB programming language (version 2024b). However, users do not need to have the MATLAB software and license to run them. The apps only need MATLAB Runtime (non-license), which is a standalone set of shared libraries enabling the execution of MATLAB-related components. The required MATLAB Runtime will be downloaded from online with the installation package.

For installation, users simply need to download the zip file from the GitHub link provided, unzip it, and double-click on the installation files. The installation package contains the installers for RAFT-Ports, along with the needed MATLAB Runtime. Users only need to select the directory where the executable file will be saved, and then the application is ready to run. Restart the computer after installation to ensure it runs efficiently.

a. Version Information

The user manual refers to the first version of the application released in 2026. The application and the user manual are uploaded in GitHub. Further revisions or updates will be posted in https://github.com/raquelpfelix/RAFT-Ports/.

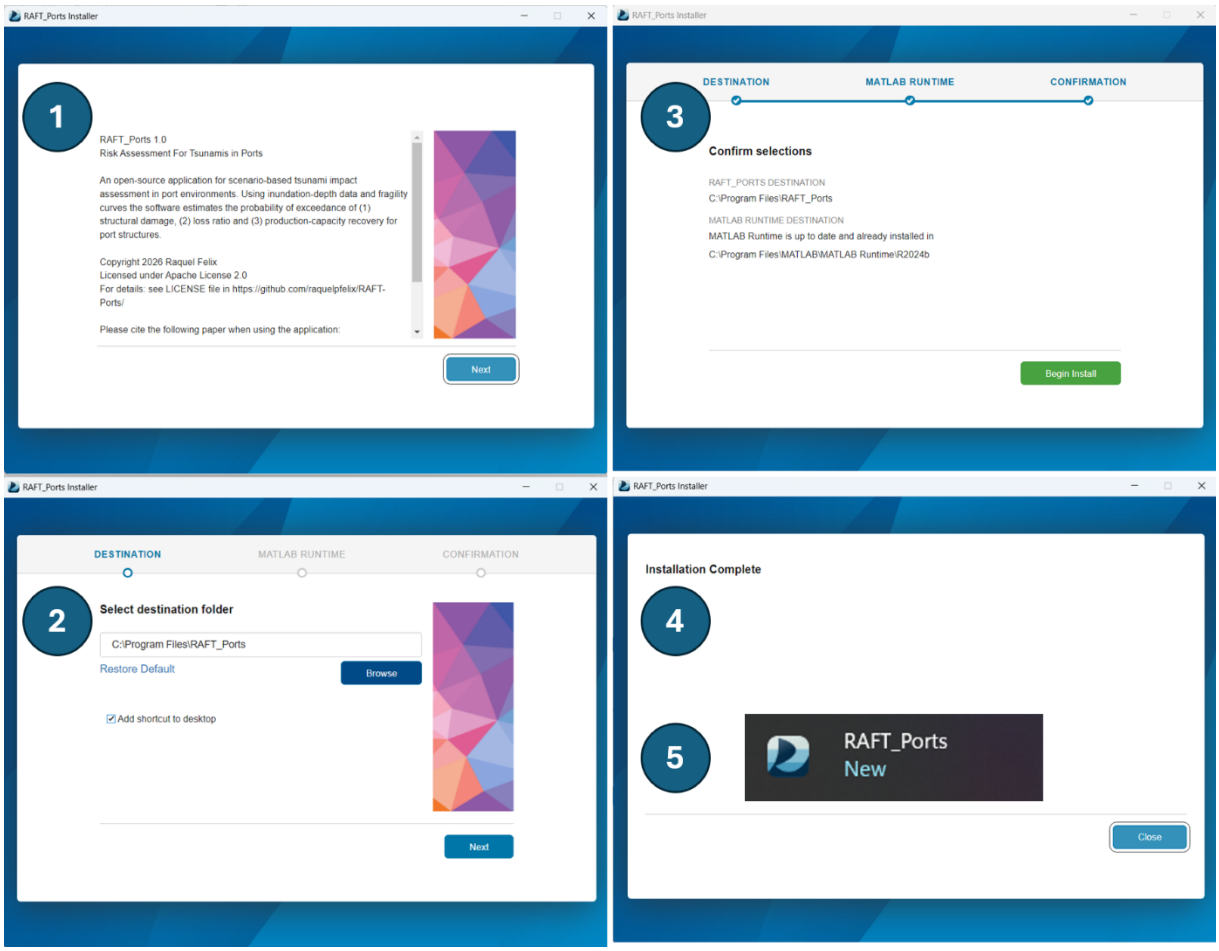
b. Disclaimer

RAFT-Ports is released under the Apache License 2.0, which permits use, modification, and redistribution of the software for both academic and commercial purposes. Users may adapt and distribute modified versions provided that the terms of the Apache License 2.0 are maintained. The full license text is included with the software distribution and is available in the GitHub repository.

Windows OS

Install RAFT-Ports as a standalone app, which does not require a MATLAB license

The installer configuration for the macOS is similar.



Installation Issues on macOS

If you encounter issues during installation, follow these steps based on your Mac's processor type:

For Apple Silicon (M1, M2, M3 chips):

You need to install Rosetta to run Intel-based applications. Follow this guide for instructions:

https://www.howtogeek.com/702982/how-to-launch-the-intel-version-of-a-universal-mac-app-on-apple-silicon/

For Intel-based Macs:

You may need to update file permissions before running the app. Open Terminal and enter the following commands:

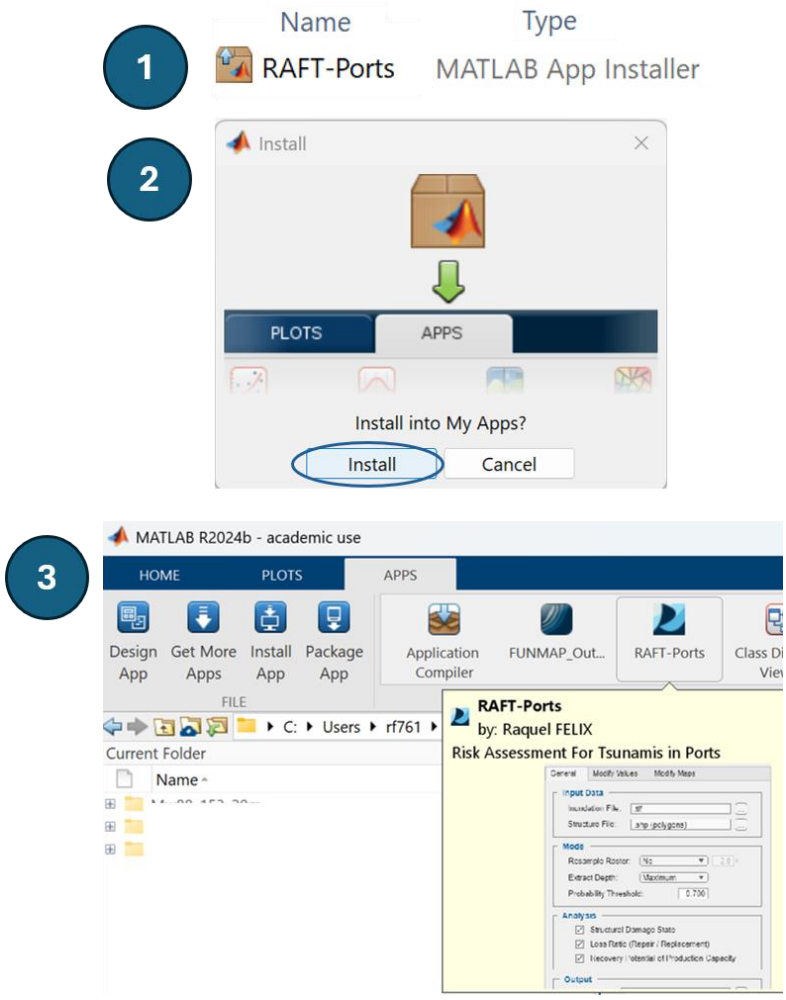
chmod -R 755 /Downloads/RAFT_Ports.app

sudo xattr -cr /Downloads/RAFT_Ports.app

Then, try running the installer again.

MATLAB APP Installer

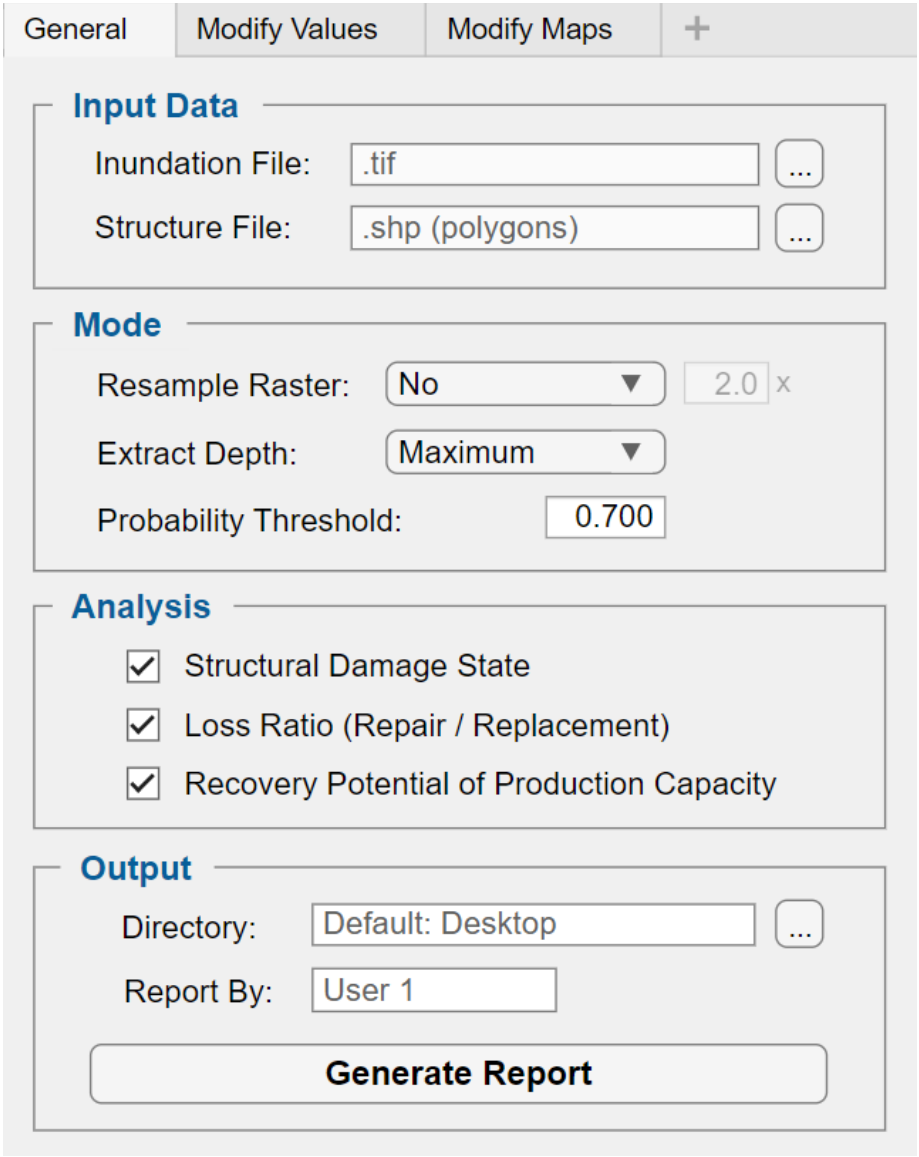
Install the app within the MATLAB software



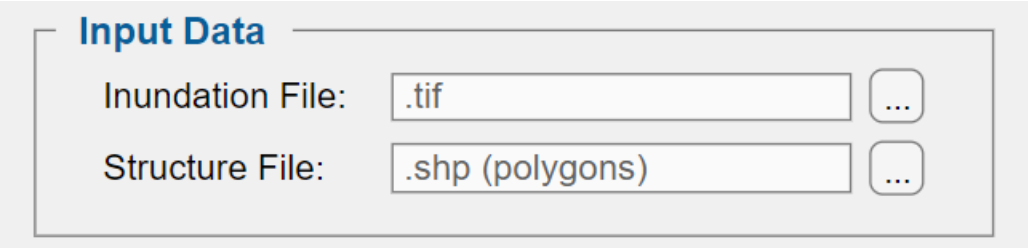
Tab 1: General

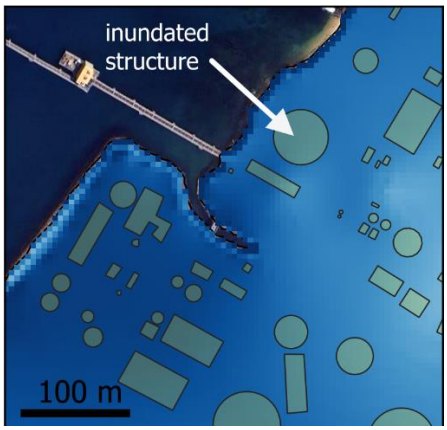
This tab contains all the functions and options required to carry out the analyses and generate the report.

Tooltips are available. Hover over the text in the application to see more details.



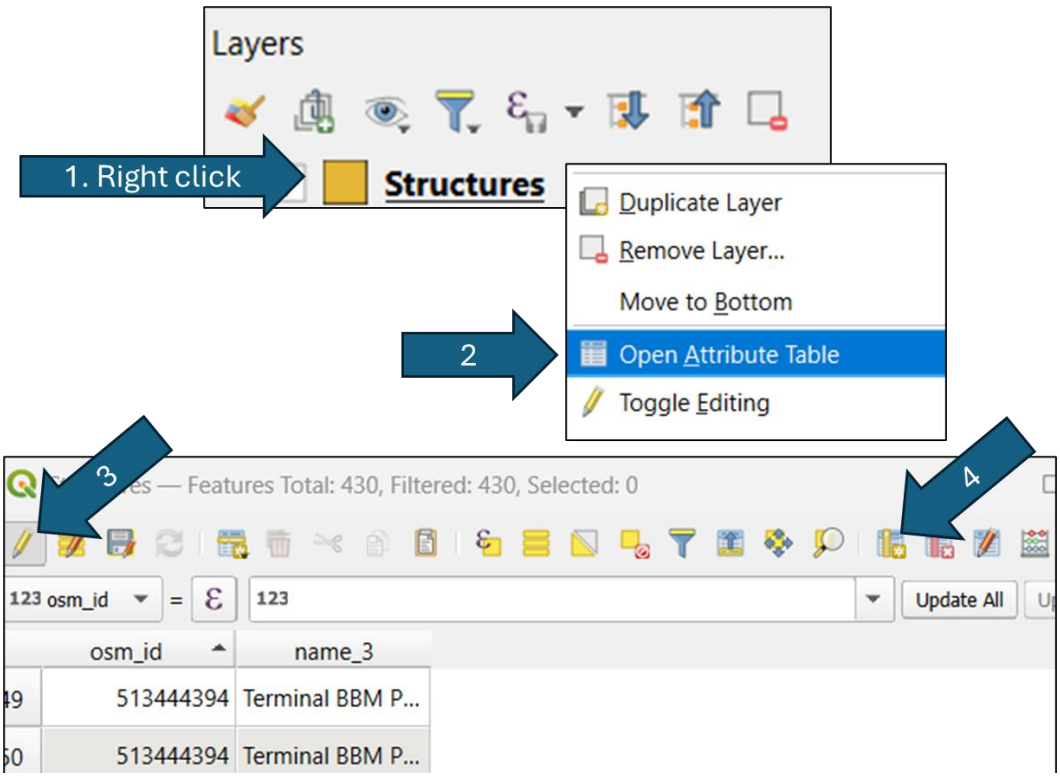
A. Input Data



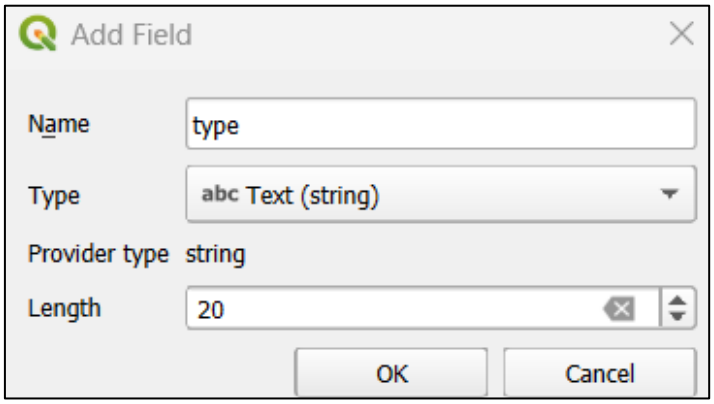
Inundation File	Click the button with three dots. A pop-up window will appear to find the file to import. Accepted format: Georeferenced .tif Assumption: The inundation raster is based on a digital surface model (DSM), so depth values exist within building footprints 
Structure File	Click the button with three dots. A pop-up window will appear to find the file to import. Format: .shp

How to add an attribute table to your shapefile (QGIS)

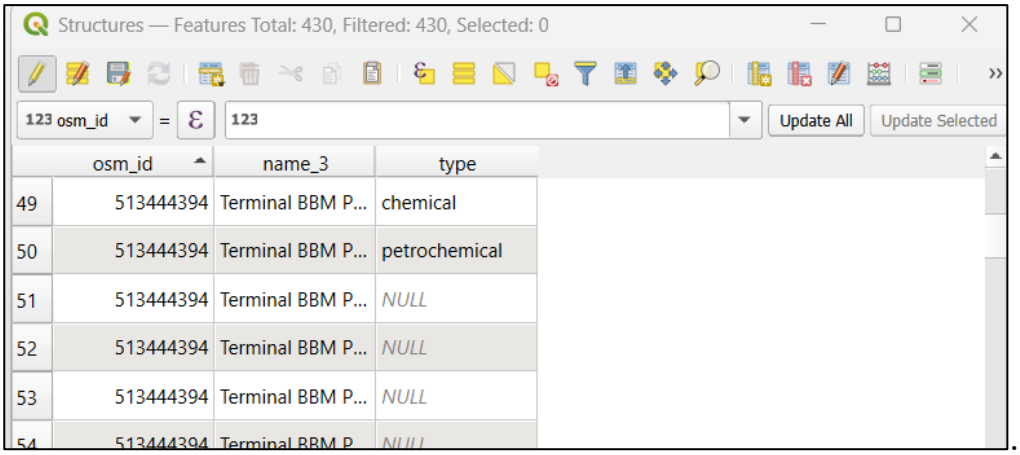
Steps 1 to 4:



Step 5: Make sure the word *type* is in small caps



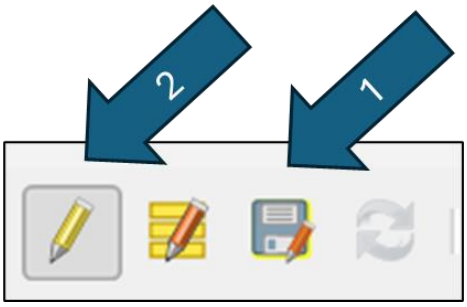
Step 6: The *type* column is now created. Simply input directly the type of industry the structure belongs to.



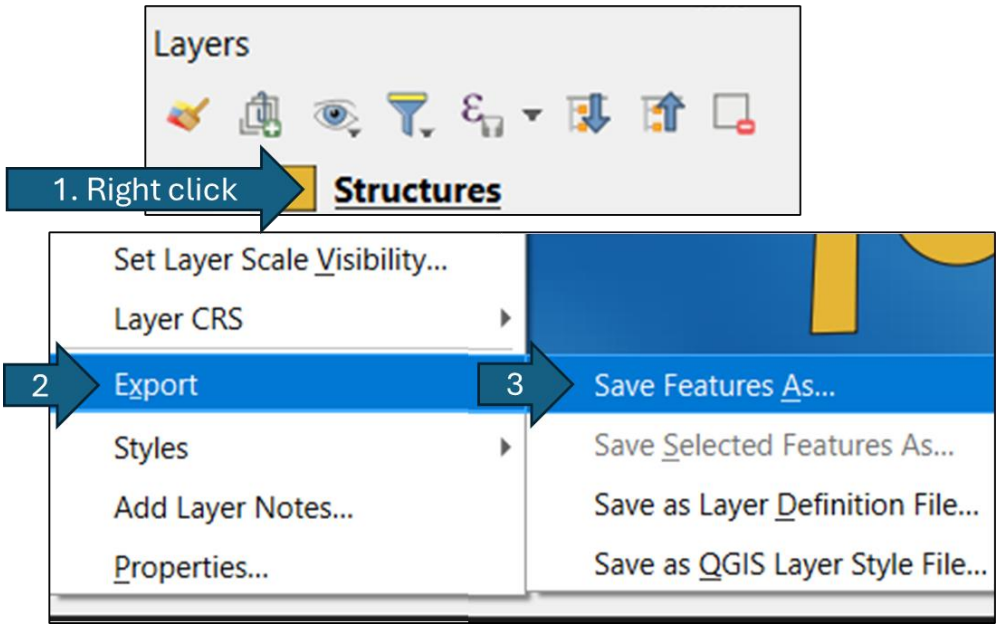
	osm_id	name_3	type
49	513444394	Terminal BBM P...	chemical
50	513444394	Terminal BBM P...	petrochemical
51	513444394	Terminal BBM P...	NULL
52	513444394	Terminal BBM P...	NULL
53	513444394	Terminal BBM P...	NULL
54	513444394	Terminal BBM P...	NULL

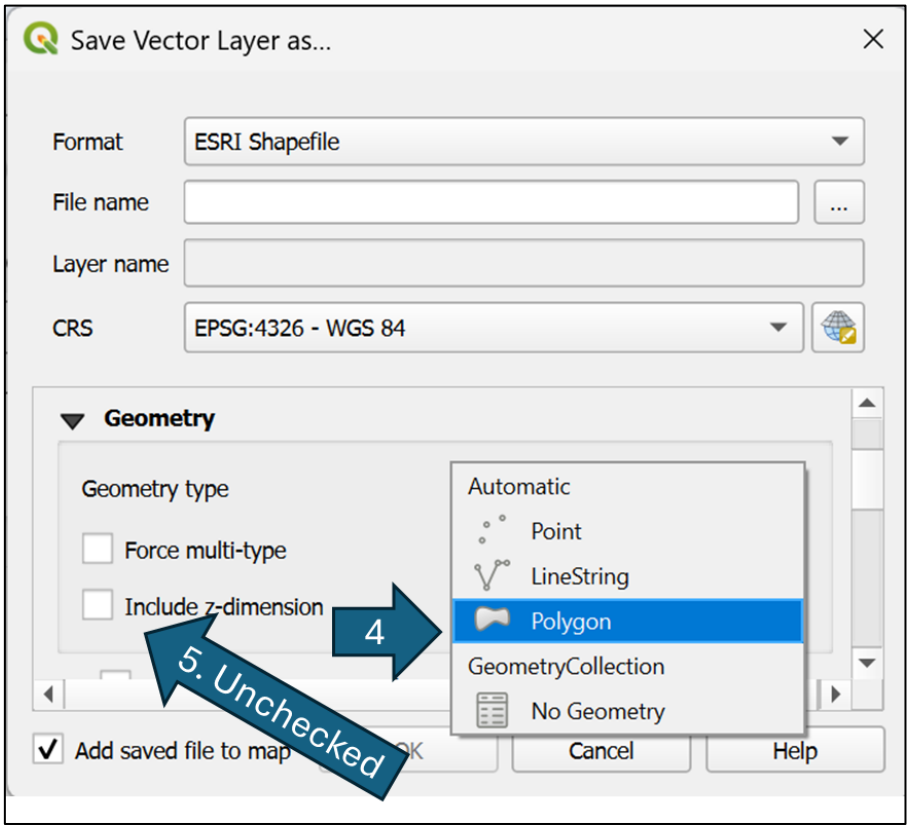
The default types in version 1 of the app are: cargo, chemical, construction, energy, food, manufacturing, petrochemical, transportation and warehouse. The app can also read other industry types if you enter them in the 'Modify Values' tab.

Step 7: Save the edits

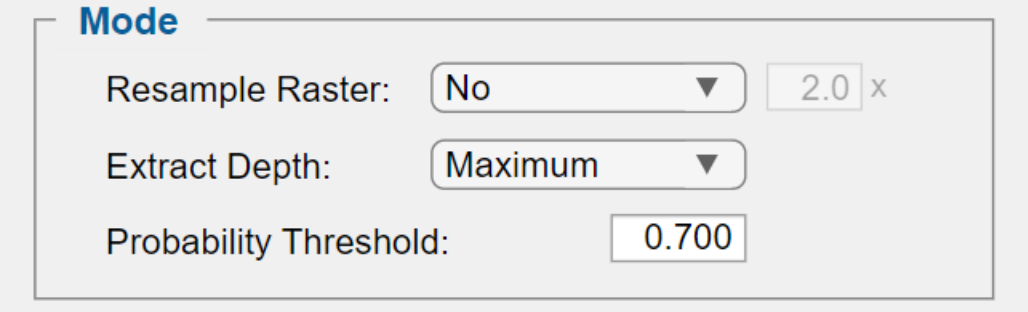


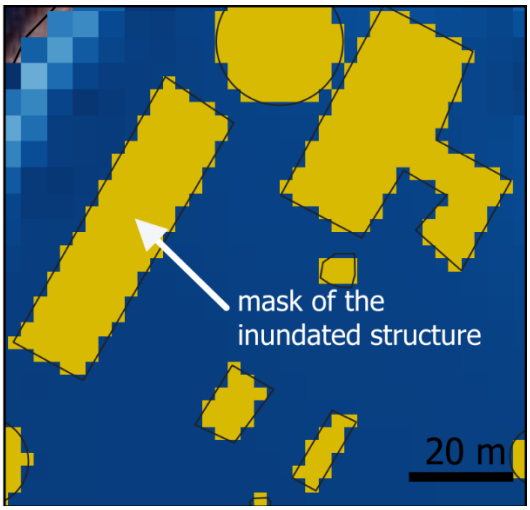
How to save the shapefile as 2D polygons

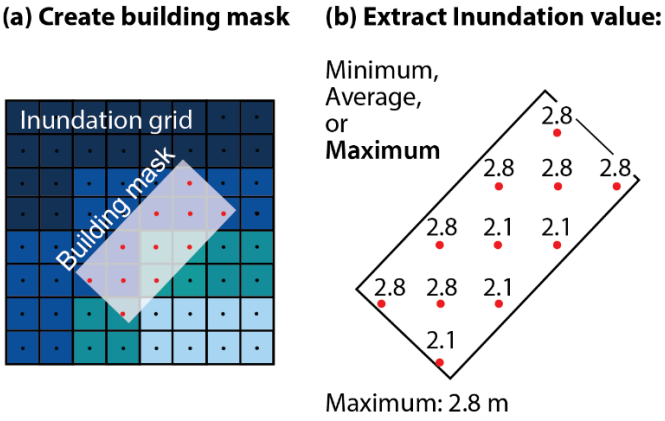




B. Mode



Resample Raster	Resample the inundation.tif file If the raster is too coarse, there might be no grid points that fall inside the structure, so no depth values will be extracted for that building. Resampling ensures there are grid points inside the building footprints. Options: No resampling, Cubic, Linear, Nearest If Cubic, Linear, or Nearest is chosen, the Resampling Factor textbox on the right side will be enabled. The map below shows the mask created by the application, saved as *Inundation_BuildingMasked.tif* in the output folder. Load it in QGIS to compare the mask boundary with the structures' polygons and decide whether the input raster needs further refinement. 

Resampling Factor	The textbox on the right side of the Resample Raster This value is used as a denominator when dividing the original inundation cell size. A higher value produces a more refined grid Example: If the grid size of the inundation.tif file is 10 m, a resampling factor of "2.0" means the inundation is refined to a grid size of 5 m.
Extract Depth	Select the depth value that will represent the inundation within a building footprint **(a) Create building mask** **(b) Extract Inundation value:** 
Probability Threshold	Select the probability threshold for displaying results Only buildings with probability values at or above the selected threshold will be shown in the report. If a value of 0.70 is selected, the interpretation will be: "A total of X buildings have a 70% probability of exceeding the threshold." This depends on the user's risk appetite.

C. Analysis



Structural Damage State	Counts the number of structures expected to experience different levels of structural damage. These levels are set to Minor, Moderate, Major and Destroyed, corresponding to DS1–DS4 in Chua et al. (2021).

Label	Damage State (DS)	Building Damage Description	Serviceability
Minor	1	Inundated but the non-structural components and roof are intact.	Functional but requires interior fixing
Moderate	2	Sustained damage on non-structural components and/or roofs.	Operational after repairs
Major	3	Sustained damage on structural components, or buckling/folding of rackings; building collapse is possible.	Not repairable
Destroyed	4	Total structural failure.	Not operational

Loss Ratio (Repair / Replacement)	Represents the proportion of repair cost relative to the full replacement cost of the structure. It provides an estimate of the degree of functional loss.
Recovery Potential of Production Capacity	Estimates the probability that a structure will reach a given level of production capacity recovery (equivalent to its full pre-earthquake capacity) after the earthquake.

The values provided in the reference tables on the following pages are the underlying data used to develop the fragility curves and are not extracted from the curves themselves. These data were provided directly by the authors of the respective studies: Chua et al. (2021), Suppasri et al. (2022), and Miki et al. (2025).

Reference: Probability of Exceedance Values for

Structural Damage States

Source: Chua et al. (2021)

https://doi.org/10.5194/nhess-21-1887-2021

Minor Structural Damage State (DS 1)

depth	*cargo*	*chem.*	*constr.*	*energy*	*food*	*manufac.*	*petro.*	*ware.*
0.5	0.8752	0.4500	0.7847	0.2383	0.7328	0.5256	0.6229	0.6250
1.0	0.9587	0.8988	0.9384	0.6271	0.9500	0.9429	0.9654	0.8950
1.5	0.9783	0.9731	0.9722	0.8194	0.9839	0.9878	0.9933	0.9567
2.0	0.9862	0.9898	0.9843	0.9013	0.9931	0.9960	0.9979	0.9775
2.5	0.9903	0.9952	0.9900	0.9401	0.9965	0.9983	0.9992	0.9866
3.0	0.9928	0.9974	0.9931	0.9606	0.9980	0.9992	0.9996	0.9912
3.5	0.9944	0.9985	0.9950	0.9726	0.9987	0.9995	0.9998	0.9939
4.0	0.9955	0.9991	0.9962	0.9800	0.9991	0.9997	0.9999	0.9955
4.5	0.9963	0.9994	0.9970	0.9849	0.9994	0.9998	0.9999	0.9966
5.0	0.9969	0.9996	0.9976	0.9882	0.9996	0.9999	1.0000	0.9973
5.5	0.9974	0.9997	0.9980	0.9907	0.9997	0.9999	1.0000	0.9979
6.0	0.9977	0.9998	0.9983	0.9924	0.9997	0.9999	1.0000	0.9983
6.5	0.9980	0.9998	0.9986	0.9937	0.9998	1.0000	1.0000	0.9986
7.0	0.9982	0.9999	0.9988	0.9947	0.9998	1.0000	1.0000	0.9988
7.5	0.9984	0.9999	0.9990	0.9955	0.9999	1.0000	1.0000	0.9990
8.0	0.9986	0.9999	0.9991	0.9962	0.9999	1.0000	1.0000	0.9991
8.5	0.9987	0.9999	0.9992	0.9967	0.9999	1.0000	1.0000	0.9992
9.0	0.9988	0.9999	0.9993	0.9971	0.9999	1.0000	1.0000	0.9993
9.5	0.9989	1.0000	0.9994	0.9975	0.9999	1.0000	1.0000	0.9994
10.0	0.9990	1.0000	0.9994	0.9978	0.9999	1.0000	1.0000	0.9995

chem. – chemical; contr. – construction; manufac. – manufacturing
petro. – petrochemical; ware. - warehouse

Moderate Structural Damage State (DS 2)

depth	*cargo*	*chem*	*constr.*	*energy*	*food*	*manufac.*	*petro.*	*ware.*
0.5	0.1201	0.0047	0.0414	0.0042	0.0076	0.0011	0.0005	0.0231
1.0	0.3020	0.0488	0.1544	0.0221	0.0536	0.0157	0.0086	0.1068
1.5	0.4598	0.1710	0.2975	0.0569	0.1537	0.0719	0.0433	0.2366
2.0	0.5798	0.3563	0.4345	0.1084	0.2932	0.1924	0.1276	0.3781
2.5	0.6667	0.5446	0.5497	0.1734	0.4416	0.3626	0.2660	0.5058
3.0	0.7303	0.6918	0.6414	0.2462	0.5716	0.5365	0.4328	0.6103
3.5	0.7771	0.7931	0.7108	0.3213	0.6759	0.6789	0.5888	0.6926
4.0	0.8134	0.8590	0.7632	0.3946	0.7538	0.7808	0.7118	0.7548
4.5	0.8412	0.9016	0.8046	0.4644	0.8113	0.8493	0.7994	0.8028
5.0	0.8637	0.9293	0.8367	0.5282	0.8540	0.8949	0.8594	0.8388
5.5	0.8816	0.9480	0.8620	0.5850	0.8855	0.9250	0.9002	0.8671
6.0	0.8958	0.9607	0.8824	0.6354	0.9087	0.9454	0.9279	0.8890
6.5	0.9075	0.9699	0.8984	0.6799	0.9263	0.9594	0.9469	0.9062
7.0	0.9171	0.9765	0.9115	0.7180	0.9398	0.9693	0.9601	0.9198
7.5	0.9257	0.9814	0.9220	0.7513	0.9500	0.9764	0.9697	0.9309
8.0	0.9327	0.9765	0.9307	0.7799	0.9581	0.9815	0.9765	0.9401
8.5	0.9387	0.9878	0.9382	0.8042	0.9647	0.9854	0.9816	0.9477
9.0	0.9440	0.9899	0.9446	0.8251	0.9699	0.9883	0.9854	0.9540
9.5	0.9485	0.9916	0.9502	0.8432	0.9741	0.9905	0.9883	0.9592
10.0	0.9525	0.9899	0.9548	0.8591	0.9776	0.9922	0.9905	0.9637

chem. – chemical; contr. – construction; manufac. – manufacturing
petro. – petrochemical; ware. - warehouse

Major Structural Damage State (DS 3)

depth	*cargo*	*chem.*	*constr.*	*energy*	*food*	*manufac.*	*petro.*	*ware.*
0.5	0.0307	0.0012	0.0171	0.0020	0.0018	0.0003	0.0001	0.0071
1.0	0.0913	0.0129	0.0684	0.0106	0.0130	0.0039	0.0024	0.0352
1.5	0.1653	0.0501	0.1469	0.0278	0.0408	0.0187	0.0122	0.0863
2.0	0.2420	0.1243	0.2382	0.0545	0.0885	0.0553	0.0381	0.1563
2.5	0.3162	0.2346	0.3315	0.0901	0.1559	0.1229	0.0895	0.2383
3.0	0.3858	0.3652	0.4195	0.1337	0.2380	0.2222	0.1718	0.3246
3.5	0.4483	0.4957	0.4990	0.1833	0.3278	0.3428	0.2813	0.4080
4.0	0.5031	0.6090	0.5657	0.2368	0.4174	0.4682	0.4021	0.4852
4.5	0.5524	0.6999	0.6239	0.2920	0.5010	0.5825	0.5197	0.5546
5.0	0.5951	0.7706	0.6733	0.3486	0.5771	0.6778	0.6252	0.6140
5.5	0.6321	0.8226	0.7153	0.4022	0.6426	0.7530	0.7105	0.6655
6.0	0.6649	0.8625	0.7503	0.4536	0.6985	0.8106	0.7783	0.7099
6.5	0.6940	0.8920	0.7798	0.5020	0.7451	0.8538	0.8297	0.7471
7.0	0.7194	0.9139	0.8051	0.5474	0.7840	0.8865	0.8688	0.7788
7.5	0.7421	0.9308	0.8270	0.5889	0.8162	0.9109	0.8975	0.8053
8.0	0.7613	0.9438	0.8456	0.6260	0.8426	0.9294	0.9192	0.8277
8.5	0.7793	0.9539	0.8608	0.6599	0.8645	0.9435	0.9359	0.8469
9.0	0.7954	0.9618	0.8746	0.6908	0.8826	0.9543	0.9485	0.8634
9.5	0.8096	0.9680	0.8858	0.7183	0.8977	0.9626	0.9583	0.8776
10.0	0.8222	0.9731	0.8960	0.7433	0.9106	0.9692	0.9660	0.8899

chem. – chemical; contr. – construction; manufac. – manufacturing
petro. – petrochemical; ware. - warehouse

Destroyed Structural Damage State (DS 4)

depth	*cargo*	*chem.*	*constr.*	*energy*	*food*	*manufac.*	*petro.*	*ware.*
0.5	0.0126	0.0004	0.0056	0.0009	0.0005	0.0001	0.0000	0.0022
1.0	0.0389	0.0042	0.0233	0.0047	0.0036	0.0016	0.0006	0.0110
1.5	0.0740	0.0171	0.0527	0.0124	0.0115	0.0077	0.0031	0.0282
2.0	0.1146	0.0448	0.0917	0.0247	0.0261	0.0234	0.0100	0.0539
2.5	0.1583	0.0918	0.1391	0.0418	0.0488	0.0540	0.0244	0.0877
3.0	0.2034	0.1589	0.1902	0.0637	0.0797	0.1040	0.0499	0.1283
3.5	0.2472	0.2436	0.2444	0.0903	0.1190	0.1747	0.0901	0.1748
4.0	0.2907	0.3381	0.2983	0.1209	0.1656	0.2631	0.1461	0.2248
4.5	0.3327	0.4334	0.3517	0.1548	0.2180	0.3610	0.2165	0.2763
5.0	0.3731	0.5236	0.4030	0.1912	0.2740	0.4605	0.2991	0.3285
5.5	0.4109	0.6039	0.4502	0.2291	0.3321	0.5531	0.3867	0.3793
6.0	0.4468	0.6727	0.4950	0.2690	0.3898	0.6349	0.4739	0.4288
6.5	0.4797	0.7305	0.5363	0.3079	0.4456	0.7039	0.5556	0.4752
7.0	0.5103	0.7774	0.5728	0.3471	0.4992	0.7605	0.6286	0.5181
7.5	0.5393	0.8160	0.6081	0.3867	0.5487	0.8060	0.6919	0.5584
8.0	0.5659	0.8475	0.6401	0.4234	0.5945	0.8426	0.7459	0.5954
8.5	0.5906	0.8727	0.6687	0.4600	0.6363	0.8714	0.7897	0.6300
9.0	0.6133	0.8928	0.6944	0.4939	0.6738	0.8944	0.8260	0.6605
9.5	0.6342	0.9092	0.7169	0.5260	0.7074	0.9127	0.8553	0.6884
10.0	0.6540	0.9227	0.7376	0.5569	0.7373	0.9274	0.8794	0.7139

chem. – chemical; contr. – construction; manufac. – manufacturing
petro. – petrochemical; ware. - warehouse

Reference: Probability of Exceedance Values for

Loss Ratio

Source: Yushi Miki following the methodology of Miki et al. (2025)

https://doi.org/10.1038/s44304-025-00127-x

depth	*cargo*	*chem.*	*constr.*	*energy*	*food*	*manufac.*	*petro.*	*ware., tran.*
0.5	0.5132	0.3233	0.4327	0.2906	0.3753	0.3335	0.3462	0.3923
1.0	0.6576	0.4818	0.5704	0.4206	0.4940	0.4810	0.4851	0.5201
1.5	0.7596	0.5623	0.6753	0.4921	0.5607	0.5391	0.5331	0.6075
2.0	0.8264	0.6535	0.7583	0.5442	0.6397	0.6166	0.6098	0.6859
2.5	0.8711	0.7446	0.8208	0.5913	0.7211	0.7078	0.7091	0.7519
3.0	0.9015	0.8193	0.8644	0.6354	0.7916	0.7911	0.8007	0.8047
3.5	0.9227	0.8729	0.8957	0.6766	0.8459	0.8550	0.8686	0.8451
4.0	0.9379	0.9095	0.9180	0.7141	0.8858	0.8995	0.9136	0.8757
4.5	0.9489	0.9344	0.9344	0.7474	0.9144	0.9297	0.9425	0.8994
5.0	0.9574	0.9516	0.9465	0.7774	0.9349	0.9498	0.9605	0.9173
5.5	0.9638	0.9634	0.9557	0.8031	0.9496	0.9634	0.9724	0.9312
6.0	0.9688	0.9720	0.9628	0.8258	0.9605	0.9727	0.9803	0.9419
6.5	0.9727	0.9781	0.9684	0.8451	0.9684	0.9793	0.9855	0.9506
7.0	0.9760	0.9827	0.9729	0.8620	0.9745	0.9840	0.9891	0.9576
7.5	0.9787	0.9861	0.9765	0.8762	0.9791	0.9874	0.9917	0.9632
8.0	0.9810	0.9887	0.9795	0.8889	0.9826	0.9900	0.9935	0.9678
8.5	0.9829	0.9907	0.9819	0.8998	0.9855	0.9919	0.9949	0.9717
9.0	0.9846	0.9923	0.9840	0.9093	0.9877	0.9934	0.9960	0.9750
9.5	0.9860	0.9935	0.9857	0.9176	0.9896	0.9946	0.9967	0.9777
10.0	0.9872	0.9945	0.9872	0.9250	0.9910	0.9955	0.9973	0.9801

contr. – construction; manufac. – manufacturing

petro. – petrochemical; ware. – warehouse; tran. - transportation

Reference: Probability of Exceedance Values for

Recovery Potential of the Production Capacity

It refers to the full production capacity equivalent to its pre-earthquake levels.

Source: Suppasri et al. (2022)

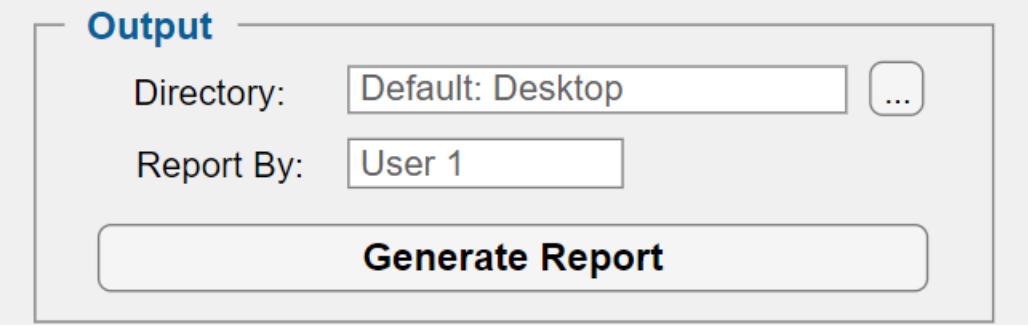
https://doi.org/10.1016/j.ijdrr.2022.103141

months	*Production Capacity*		
	No Inundation	**<3m depth**	**≥3m depth**
0.5	0.015	0	0
1.0	0.070	0.005	0.005
1.5	0.215	0.020	0.015
2.0	0.460	0.045	0.030
2.5	0.720	0.090	0.060
3.0	0.900	0.160	0.110
3.5	0.980	0.255	0.180
4.0	1.000	0.380	0.275
4.5	1.000	0.515	0.385
5.0	1.000	0.650	0.505
5.5	1.000	0.765	0.625
6.0	1.000	0.855	0.735
6.5	1.000	0.920	0.825
7.0	1.000	0.960	0.890
7.5	1.000	0.980	0.940
8.0	1.000	0.995	0.965
8.5	1.000	1.000	0.985
9.0	1.000	1.000	0.995
9.5	1.000	1.000	1.000
10	1.000	1.000	1.000

Months – refers to the month number after the earthquake.

Depth – refers to the reference inundation depth at the inundated building.

D. Output

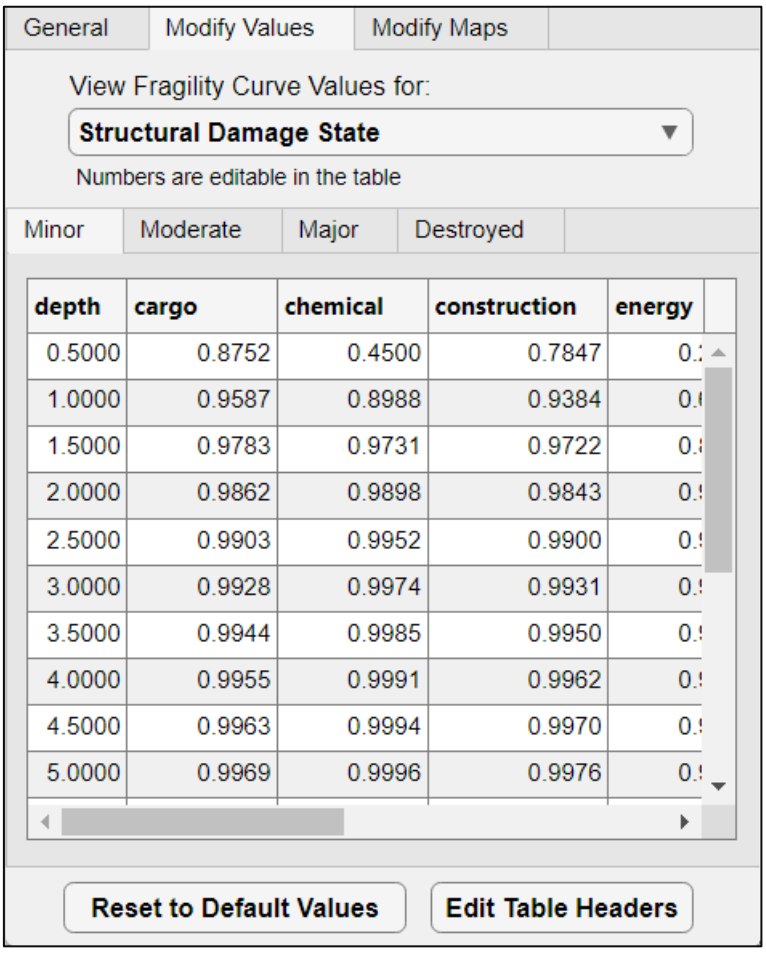


Directory	Directory Folder location for saving files. To change the directory, click the button with three dots. A new folder named 'OUTPUT_REPORT/' is created at the specified path. If no directory is selected, the default location is set to the Desktop.
Report By (Optional)	Include the name of the user on the front page of the generated PDF report.
Generate Report	Press this button to create the report. If this is not the first time you are generating the files, make sure that the previous PDF and any other files produced by the application are closed to avoid errors.

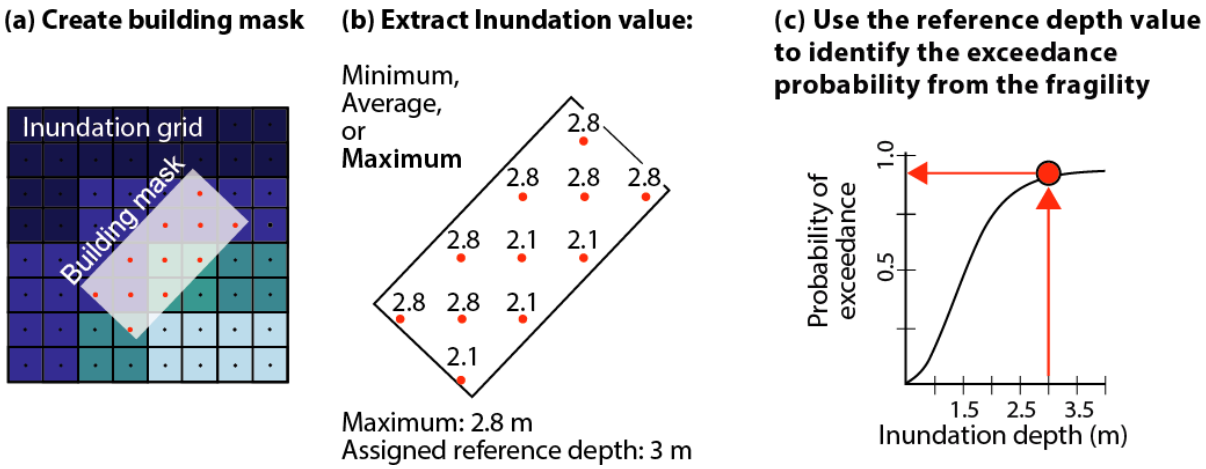
Tab 2: Modify Values

This section lists all the probability-of-exceedance values used in the analysis for reference depths ranging from 0.5 m to 9.5 m, at 0.5 m intervals.

To change the values, edit the values directly within the cells of the table.



These depth values in the tables are referred to as the *reference depths* for each 0.5 m inundation-depth interval.



Example:

A reference depth of **3 m** corresponds to structures with inundation depths **>2.5 m up to ≤3.0 m**.

This means that if a structure is inundated by **2.8 m**, the probability-of-exceedance value associated with the **3 m** reference depth will be used.

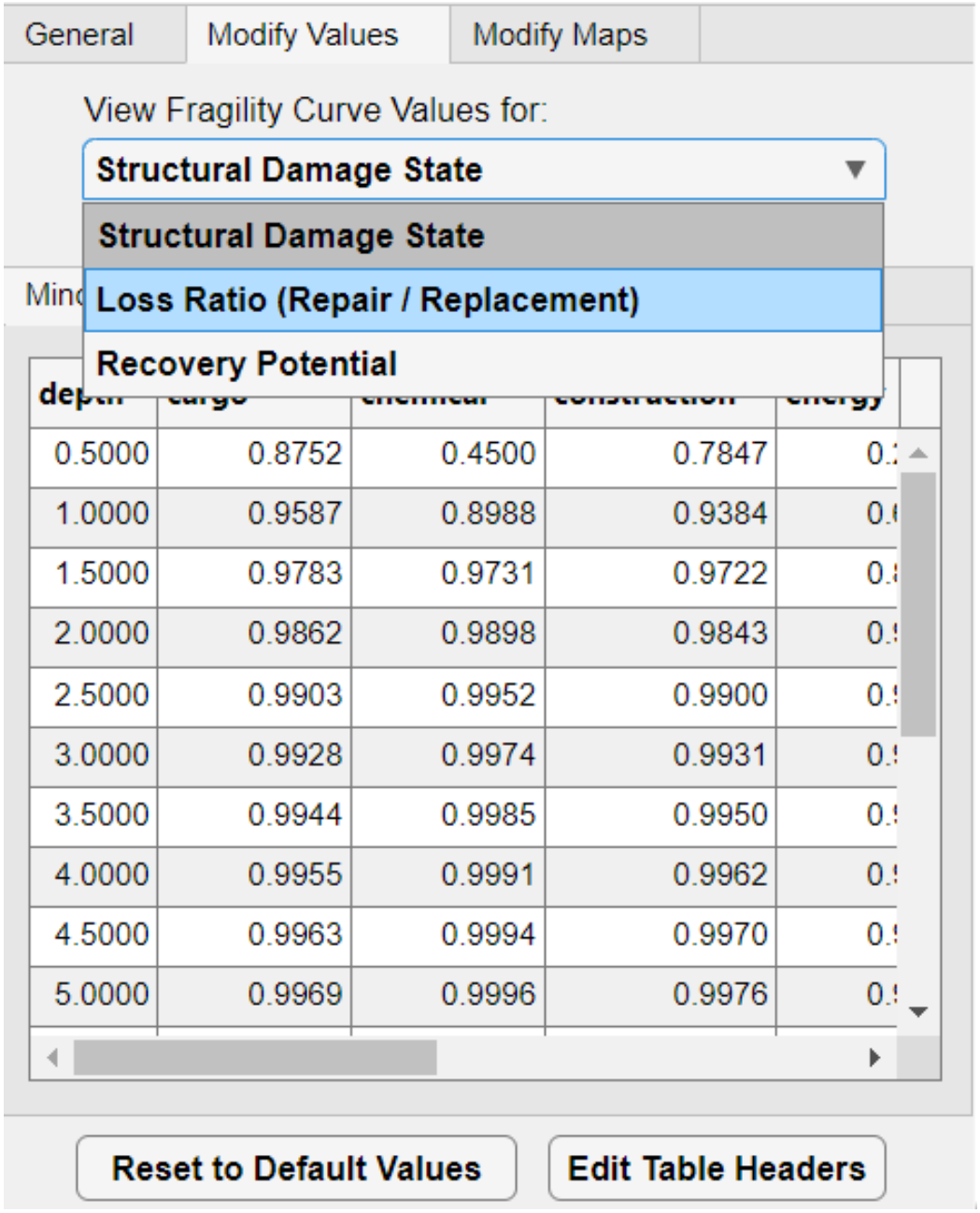
If you wish to use values from alternative fragility curves:

1. Check whether the study provides an open-access database or supplementary data.

2. Contact the corresponding author to request the underlying data.

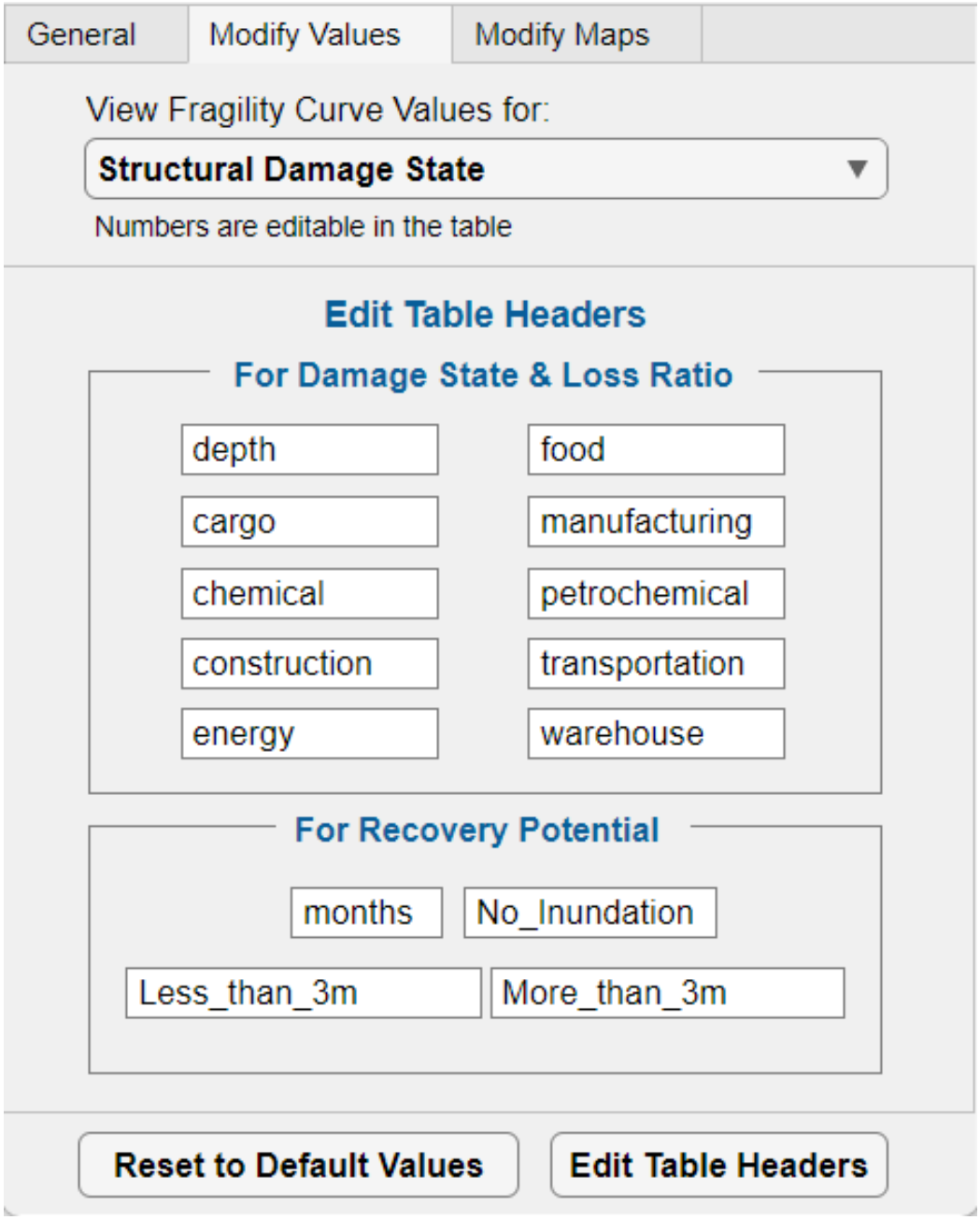
3. If permitted under the publication's licence, extract data from published fragility curve figures using graph digitisation tools:
   3.1. Upload an image of the fragility curve to a graph digitisation tool.
   3.2. Calibrate the graph by defining known reference values on the x- and y-axes.
   3.3. Select points along the fragility curve.
   3.4. The tool will return the corresponding x-axis value (e.g., inundation depth) and probability-of-exceedance value (y-axis) for each selected point.
   3.5. Repeat this process for multiple points along the curve to generate a dataset that can be used to develop a reference table for RAFT-Ports.

   Examples of online graph digitisation tools include:
   - PlotDigitizer (plotdigitizer.com)
   - GraphReader (graphreader.com)



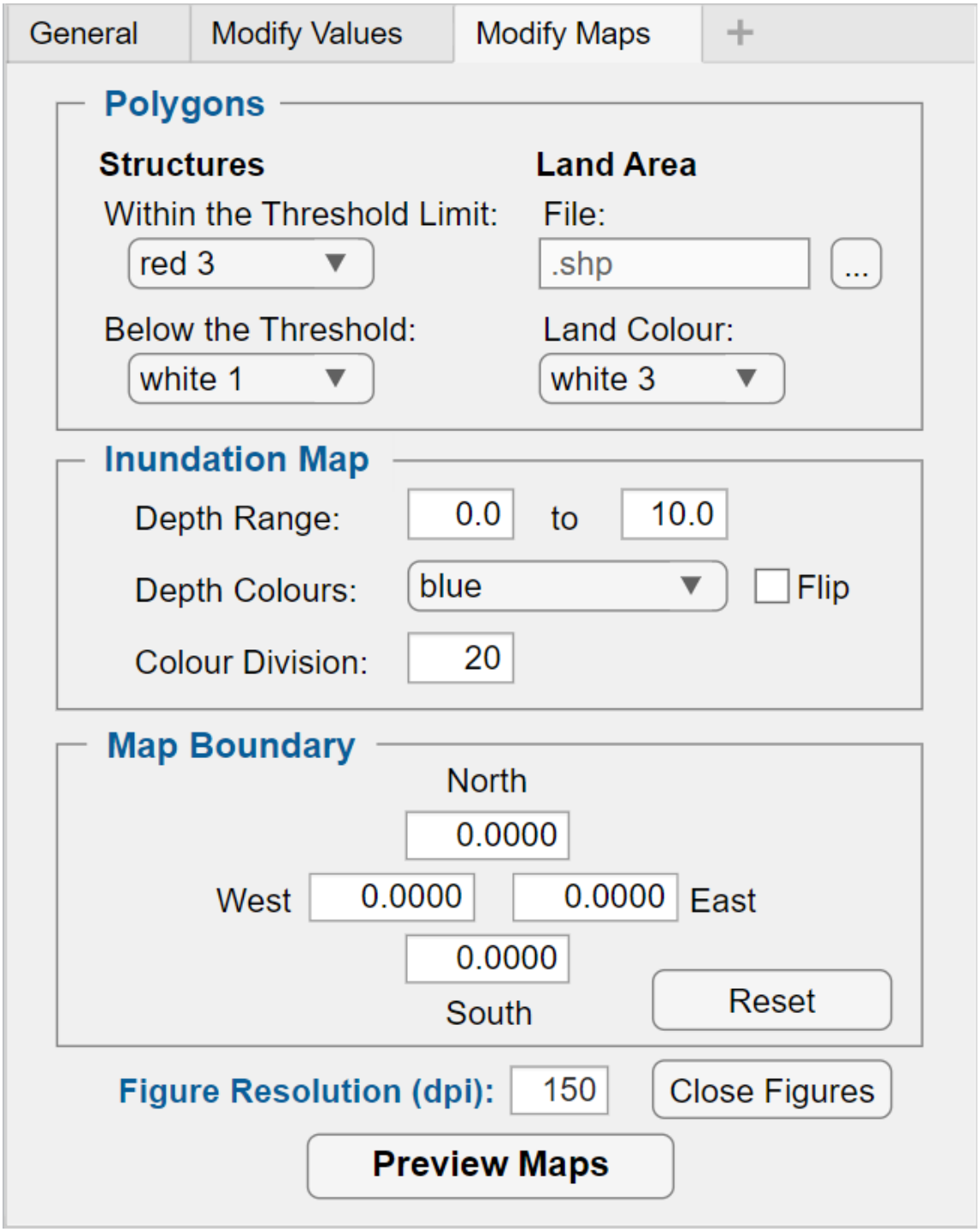
Dropdown List	Select which of the three risk-metric tables to display.
Reset to Default Values	Removes all user edits to the numeric values and restores the default values.
Edit Table Headers	To show the page where you can edit the column names (for example, to change the industry type).

Preview of the page shown when the ‘Edit Table Headers’ button is selected.

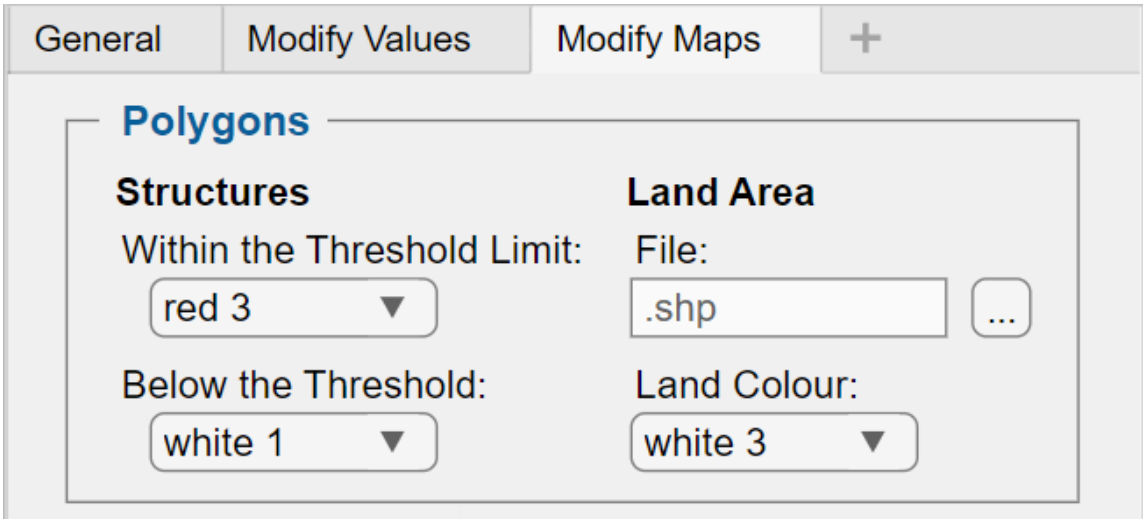


Tab 3: Modify Maps

Optional visualisation settings for the structures and the inundation map. If left unchanged, the application will still run the analysis.

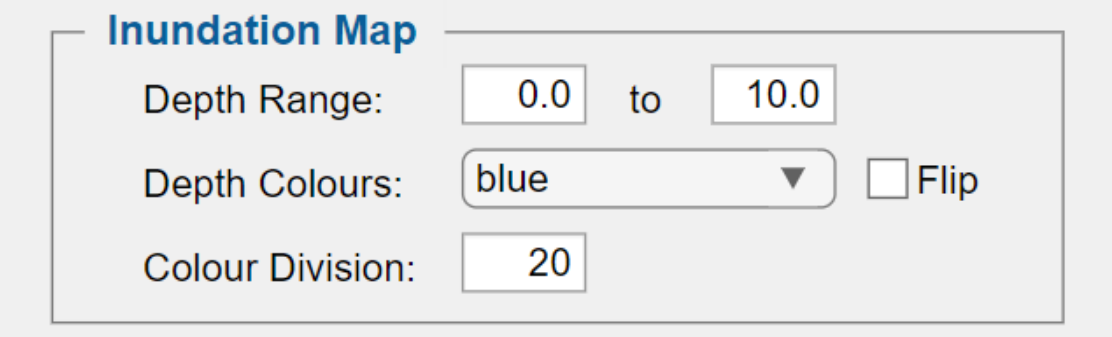


A. Polygons



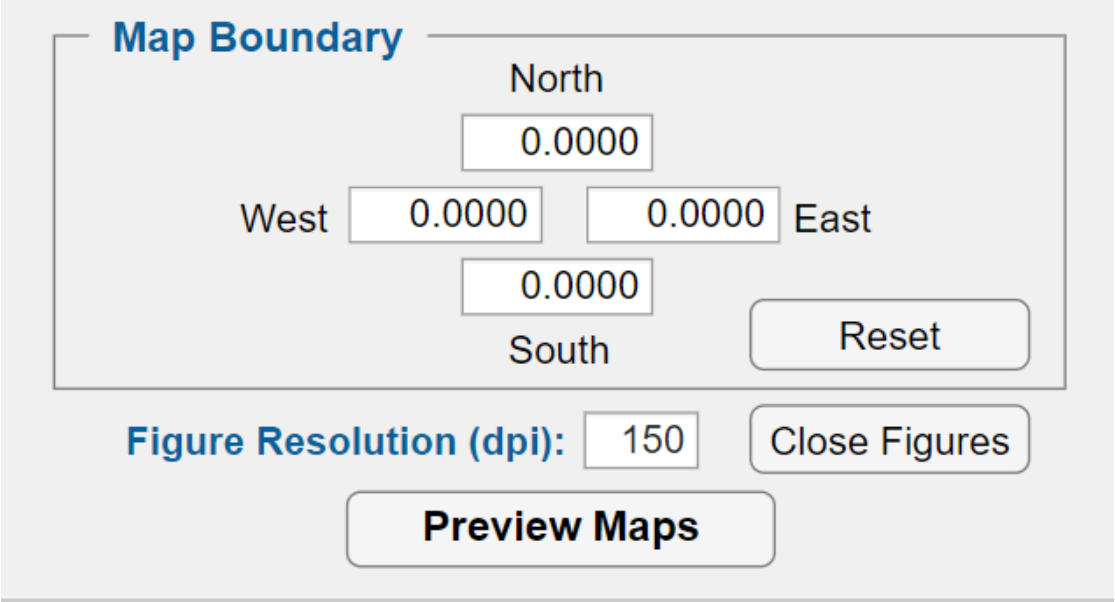
Structures	
Within the Threshold	Change the polygon colour of structures that meet or exceed the selected probability-of-exceedance threshold. For example, structures with values of 0.70 or above. Default colour**:** Red
Below the Theshold	Change the polygon colour of structures with values below the selected probability threshold. Default colour**:** White
Land Area	
File	Upload a polygon shapefile representing the land area. This is useful when the uploaded *.tif* file includes offshore wave-height values, allowing you to clearly see the boundary between land and sea.
Land Colour	Adjust the background colour of the land area so that non-inundated regions appear visually distinct from the water body.

B. Inundation Map



Depth Range	Input both the minimum and maximum values to define the colourbar range to be displayed on the map.
Depth Colours	Assign colours to wave heights using built-in and cbrewer2 package options. Visit MATLAB colourmap and cbrewer2 package websites for visuals.
Flip Checkbox	Reverses the colourmap sequence.
Colour Division	Defines the colour transition refinement. A higher value results in smoother transitions. Auto-enabled when a cbrewer2 colourmap is selected.

C. Map Boundary



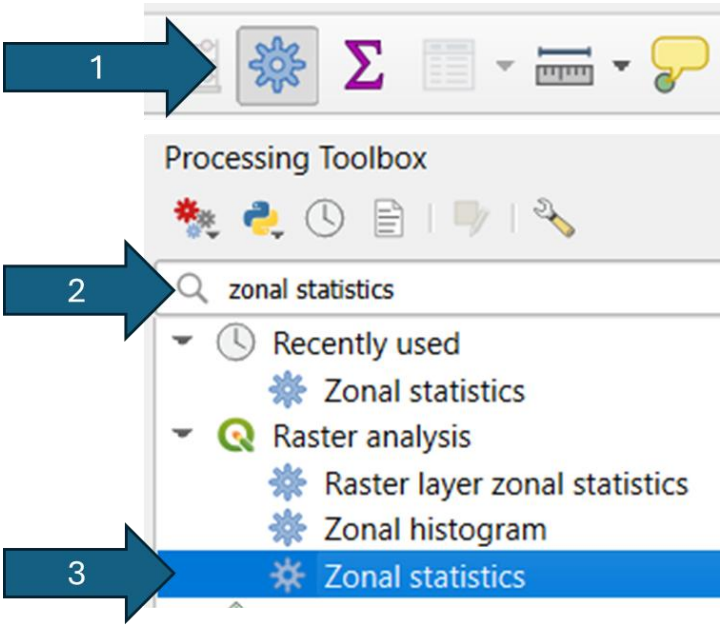
Map Boundary	Adjust the longitude and latitude limits of the map. It is automatically set to match the boundary limits of the imported .tif file
Reset Button	Reset the boundary limits to the original bounds of the imported .*tif* file.
Figure Resolution (dpi)	Set the figure quality (applies to all figures, including bar plots) Higher DPI values produce sharper imagesDefault is 150. See it higher for better resolution Units: Dots per inch
Close Figures	Close all currently open figures. Note: Existing figure windows are automatically closed before new ones are generated when clicking the 'Generate' button.
Preview Maps	View the maps after adjusting any map-making settings.

Code validation: Comparing results from QGIS & Excel

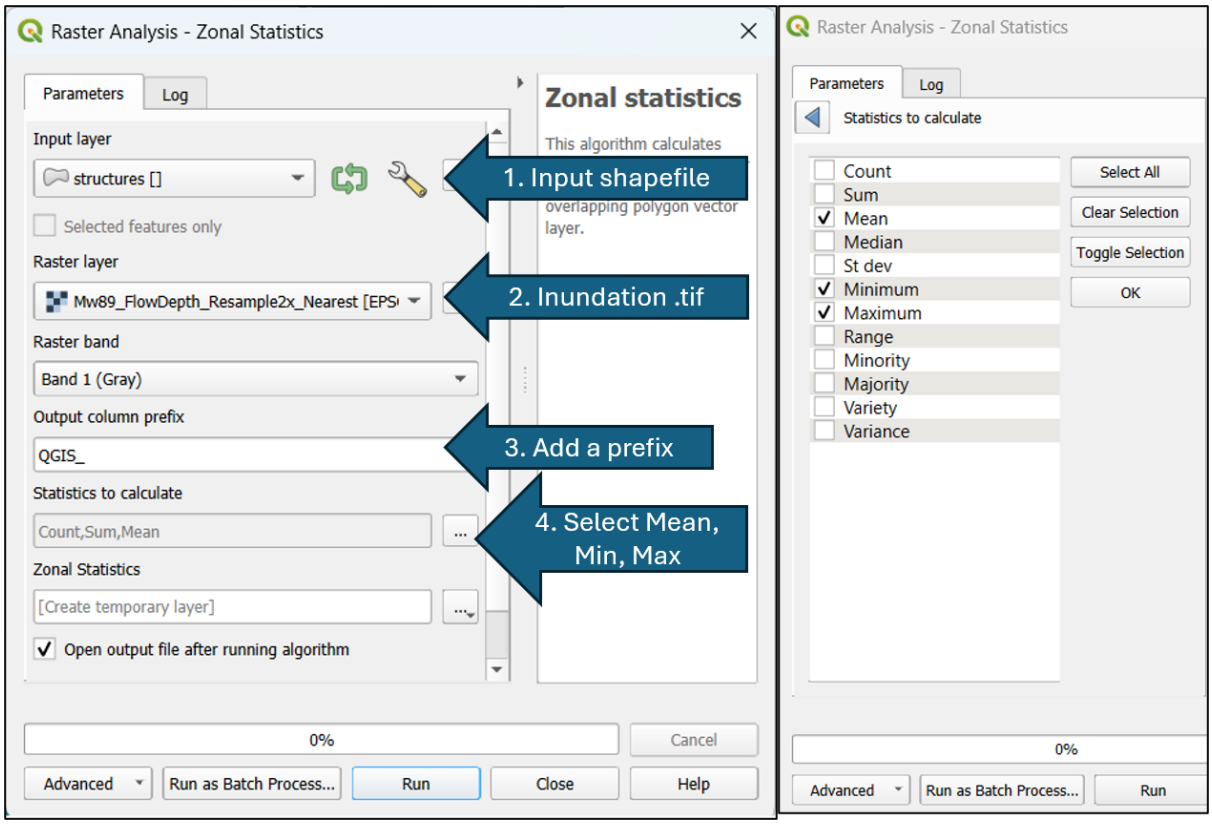
The full results of the manual comparison between the values obtained from the QGIS–Excel calculations and those generated by the RAFT-Ports App are available in the file **Validation_QGIS_vs_RAFT-Ports.xls**, included in the RAFT-Ports package on GitHub at github.com/raquelpfelix/RAFT-Ports.

Steps on using Zonal Statistics in QGIS:

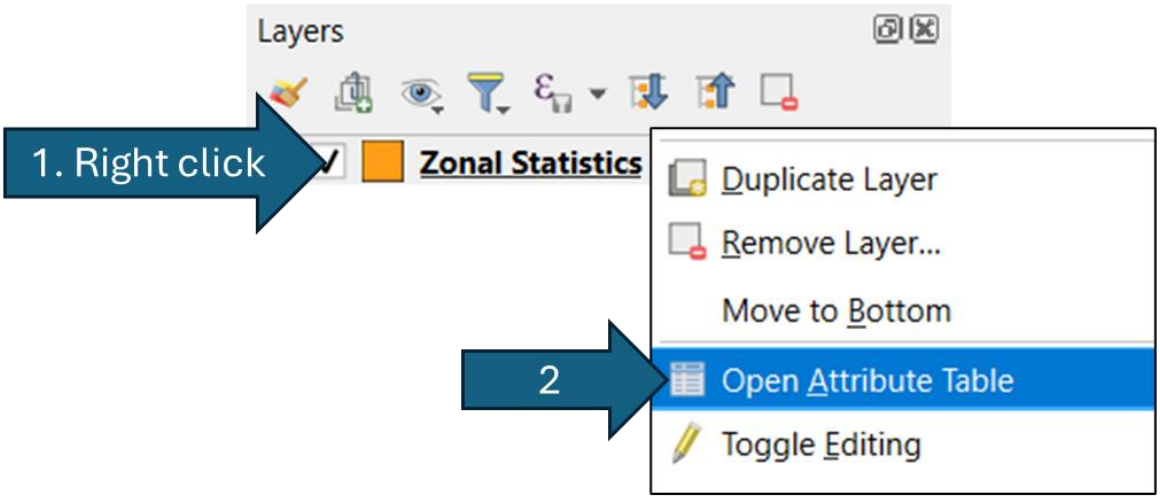
- Open QGIS
- Look for the 'Zonal Statistics' tool > For statistics to calculate, select 'Mean', 'Minimum', and 'Maximum'
- Export the created layer as .csv file



QGIS: Extract inundation depth values using Zonal Statistics



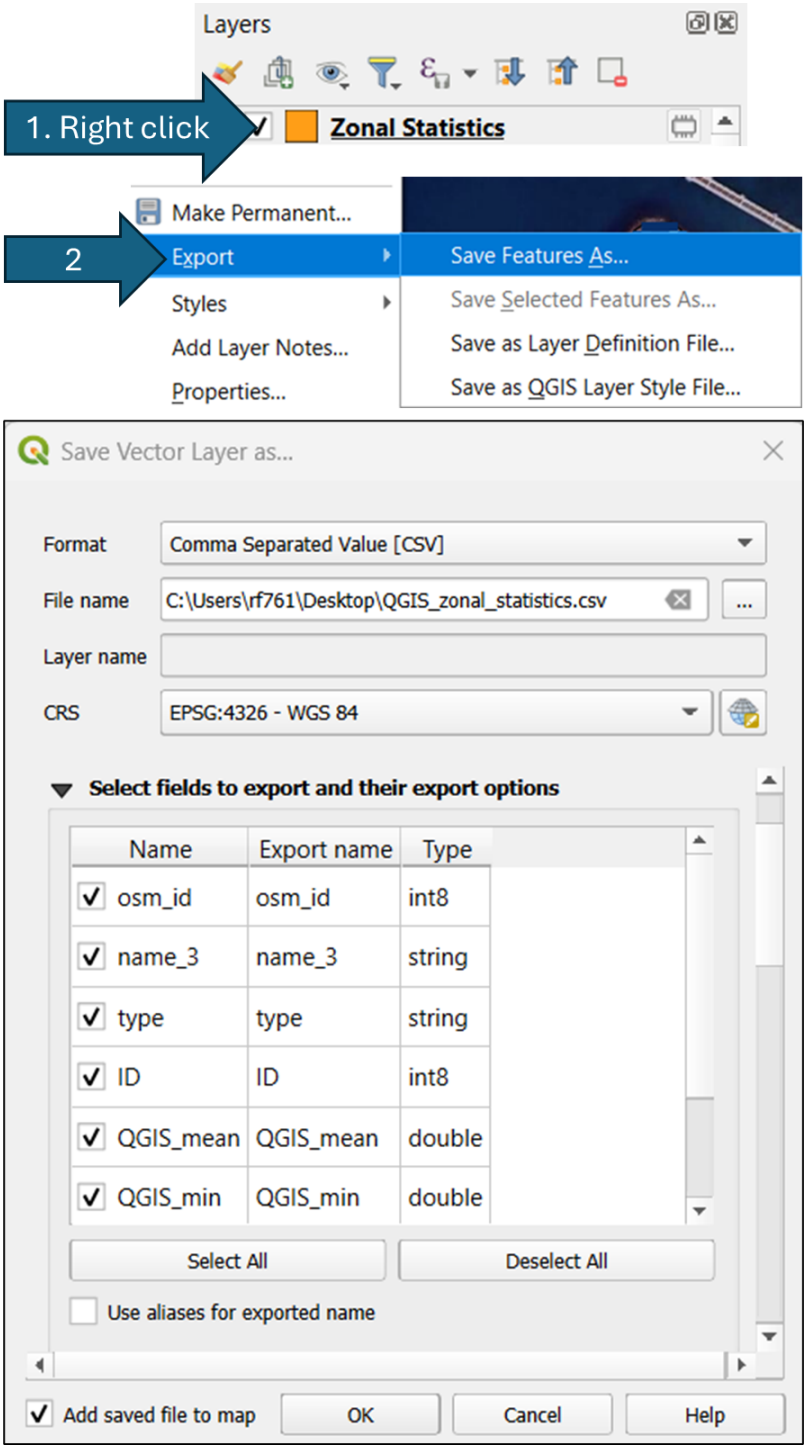
Attribute table of the created layer



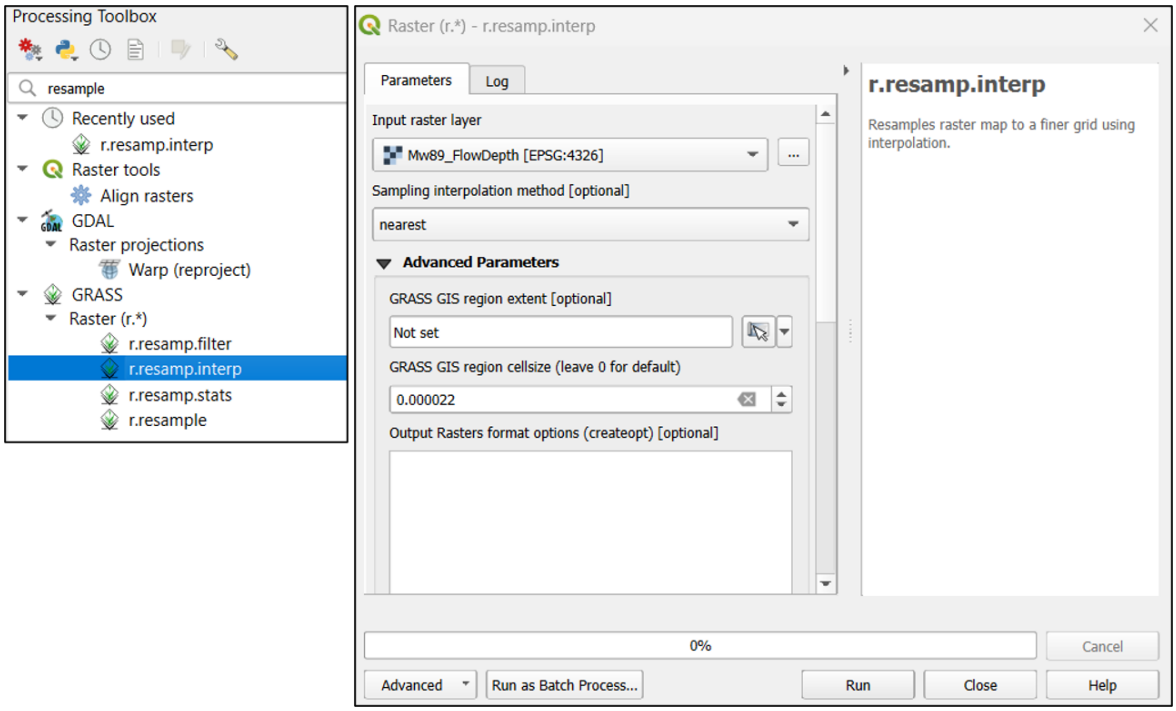
Zonal Statistics — Features Total: 430, Filtered: 430, Selecte...

	osm_id	name_3	type	ID	QGIS_mean	QGIS_min	QGIS_max
1	513444927	PT. Mits...	chemical	1	0.9984465...	0.973973...	1.012255...
2	513444927	PT. Mits...	chemical	2	1.2763773...	1.264830...	1.285918...
3	513444927	PT. Mits...	chemical	3	1.2819879...	1.276999...	1.286976...
4	513444927	PT. Mits...	chemical	4	1.2673902...	1.261962...	1.273890...
5	513444394	Termina...	petrochemical	5	1.6764036...	1.642996...	1.711966...

Saving the output layer as .csv



If you choose to resample the inundation raster in RAFT-Ports, you must also resample the original inundation raster in QGIS before performing the validation. When creating the resampled raster in QGIS, use the **Updated Resolution** values listed on the front page of the **Report.pdf**. This ensures that both datasets have the same spatial resolution and that the validation results are consistent and comparable.



Excel: Formulas for calculating the Probability of Exceedance

You can find all the formula in the **'Probability of Exceedance' tab of the Validation_QGIS_vs_RAFT-Ports.xls**

This tab of the Excel file focuses on the results obtained when the maximum depth is selected as the inundation value to be extracted for each structure. The analysis is based on the outputs generated when running the Mw 8.9 scenario, which is included as the test case in the installer package on GitHub.

Example code at G9 for calculating the probability of exceedance for the minor structural damage:

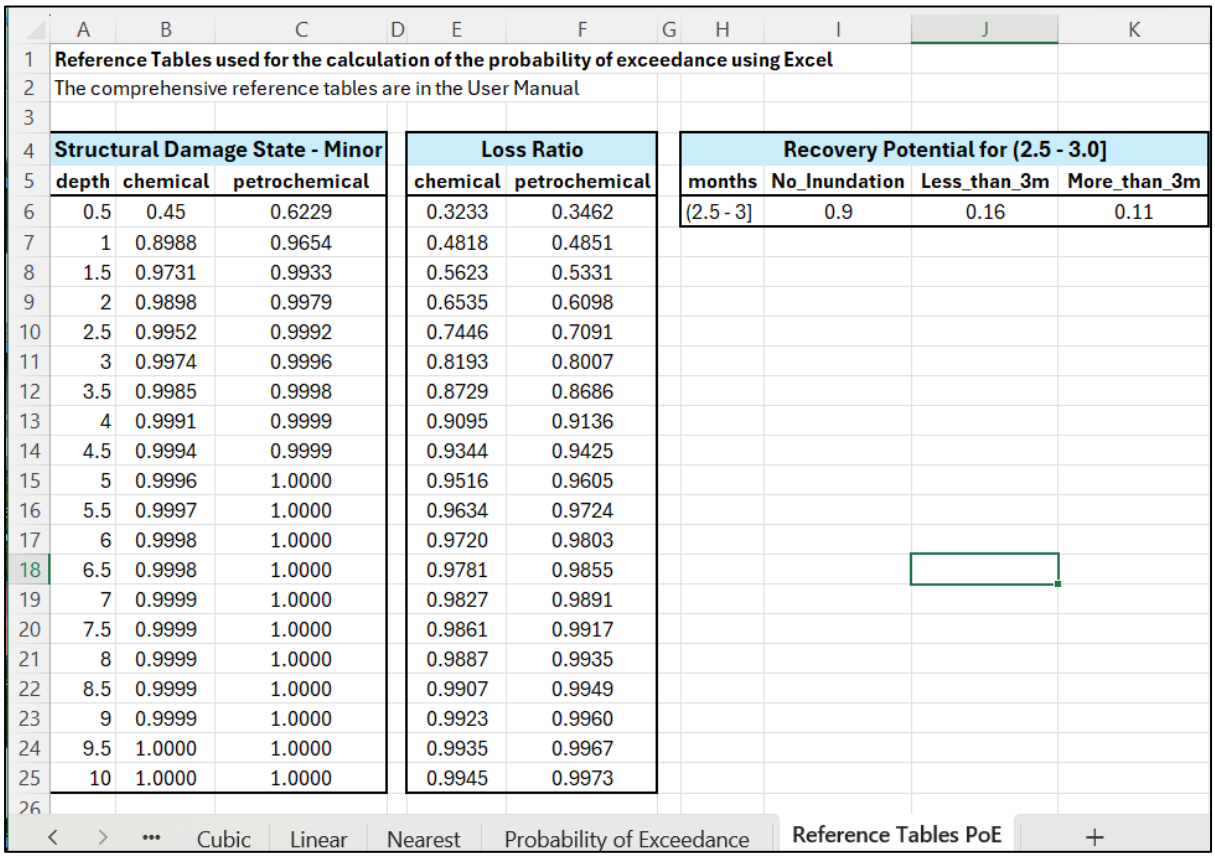
=IF(D9=0,0,INDEX('Reference Tables PoE'!$B$7:$C$26,MATCH(D9,'Reference Tables PoE'!$A$7:$A$26,0),MATCH(B9,'Reference Tables PoE'!$B$6:$C$6,0)))

Tab Preview:

Validation of the Assignment of the Probability of Exceedance

Mw 8.9 Scenario
Depth Extracted: Maximum
Resampling Method: Nearest
Resampling Factor: 2

				Probability of Exceedance								
				Minor Structural Damage			Loss Ratio			Recovery Potential for Month: (2.5-3.0]		
name_3	type	ID	DepthGroup (RA	RAFT-Ports	Excel Calculation	Mismatched? 1=Y	RAFT-Ports	Excel Calculation	Mismatched? 1=Y	RAFT-Ports	Excel Calculation	Mismatched? 1=Y
PT. Mitsu	chemical	1	1	0.8988	0.8988	0	0.4818	0.4818	0	0.16	0.16	0
PT. Mitsu	chemical	2	1.5	0.9731	0.9731	0	0.5623	0.5623	0	0.16	0.16	0
PT. Mitsu	chemical	3	1.5	0.9731	0.9731	0	0.5623	0.5623	0	0.16	0.16	0
PT. Mitsu	chemical	4	1.5	0.9731	0.9731	0	0.5623	0.5623	0	0.16	0.16	0
Terminal	petrochemical	5	2	0.9979	0.9979	0	0.6098	0.6098	0	0.16	0.16	0
Terminal	petrochemical	6	2	0.9979	0.9979	0	0.6098	0.6098	0	0.16	0.16	0
Terminal	petrochemical	7	2	0.9979	0.9979	0	0.6098	0.6098	0	0.16	0.16	0
Terminal	petrochemical	8	2.5	0.9992	0.9992	0	0.7091	0.7091	0	0.16	0.16	0
Terminal	petrochemical	9	2.5	0.9992	0.9992	0	0.7091	0.7091	0	0.16	0.16	0



Reference Tables used for the calculation of the probability of exceedance using Excel

The comprehensive reference tables are in the User Manual

Structural Damage State - Minor		
depth	chemical	petrochemical
0.5	0.45	0.6229
1	0.8988	0.9654
1.5	0.9731	0.9933
2	0.9898	0.9979
2.5	0.9952	0.9992
3	0.9974	0.9996
3.5	0.9985	0.9998
4	0.9991	0.9999
4.5	0.9994	0.9999
5	0.9996	1.0000
5.5	0.9997	1.0000
6	0.9998	1.0000
6.5	0.9998	1.0000
7	0.9999	1.0000
7.5	0.9999	1.0000
8	0.9999	1.0000
8.5	0.9999	1.0000
9	0.9999	1.0000
9.5	1.0000	1.0000
10	1.0000	1.0000

Loss Ratio	
chemical	petrochemical
0.3233	0.3462
0.4818	0.4851
0.5623	0.5331
0.6535	0.6098
0.7446	0.7091
0.8193	0.8007
0.8729	0.8686
0.9095	0.9136
0.9344	0.9425
0.9516	0.9605
0.9634	0.9724
0.9720	0.9803
0.9781	0.9855
0.9827	0.9891
0.9861	0.9917
0.9887	0.9935
0.9907	0.9949
0.9923	0.9960
0.9935	0.9967
0.9945	0.9973

Recovery Potential for (2.5 - 3.0]			
months	No_Inundation	Less_than_3m	More_than_3m
(2.5 - 3]	0.9	0.16	0.11

Cubic | Linear | Nearest | Probability of Exceedance | Reference Tables PoE

Explanation for the formula:

=IF(D9=0,0,INDEX('Reference Tables PoE'!$B$6:$C$25,MATCH(D9,'Reference Tables PoE'!$A$6:$A$25,0),MATCH(B9,'Reference Tables PoE'!$B$5:$C$5,0)))

In 'Probability of Exceedance' tab

B9: Industry type of the selected structure

D9: DepthGroup

In 'Reference Tables PoE' tab

B6 to C25: Reference table for the probability of exceedance of the Minor Structural damage state

1. Check whether D9 (assigned DepthGroup) is equal to 0.

If D9 = 0, the formula returns a value of 0.

2. If D9 is not 0, identify the correct row using:

MATCH(D9,'Reference Tables PoE'!$A$6:$A$25,0)

This matches the structure's DepthGroup (cell D9 in 'Probability of Exceedance' tab)

to the corresponding row in the reference table, using the Depth Group values listed in cells A6 to A25 of the ‘Reference Tables PoE’ tab.

3. Identify the correct column using:

MATCH(B9,'Reference Tables PoE'!$B$5:$C$5,0)))

This finds which among the columns B5 and C5 in the reference table matches the industry type listed in B9 cell of the ‘Probability of Exceedance’ tab. In the example, the B9 cell has ‘chemical’ as its industry type. It matches with the B5 cell in the ‘Reference Tables PoE’ tab.

4. Extract the probability value using:

INDEX('Reference Tables PoE'!$B$6:$C$25, row_number, column_number)

This uses the row and column numbers returned by the two MATCH functions to extract the probability of exceedance for the specific combination of DepthGroup and industry type.

List of outputs generated by RAFT-Ports

The application will save all these files in the 'OUTPUT_REPORT' folder.

- Report
- 0_Polygon_Masterfile
- Inundation_Overview
- InundationDetailedCount
- LossRatio_DetailedIndustry_CountPerDepth
- LossRatio_Overview
- RecoveryPotential_CountPerDepth_3DepthDivisions
- RecoveryPotential_CountPerDepth_DetailedDepth
- RecoveryPotential_Overview
- RecoveryPotential_SummaryPage
- ReferenceTable_LossRatio
- ReferenceTable_RecoveryPotential
- ReferenceTable_StructuralDamage_destroyed
- ReferenceTable_StructuralDamage_major
- ReferenceTable_StructuralDamage_minor
- ReferenceTable_StructuralDamage_moderate
- StructDamage_Detailed_AllTypes
- StructDamageState_destroyed_CountperDepth
- StructDamageState_major_CountperDepth
- StructDamageState_minor_CountperDepth
- StructDamageState_moderate_CountperDepth
- StructDamageState_Overview
- StructDamageState_Overview_CountPerDepth
- BarPlot_destroyed_TypeClassified
- BarPlot_inundation
- BarPlot_LossRatio
- BarPlot_major_TypeClassified
- BarPlot_minor_TypeClassified
- BarPlot_moderate_TypeClassified
- BarPlot_RecoveryPotential
- BarPlot_StructuralDamageState
- Map_Inundation
- Map_LossRatio
- Map_RecoveryPotential_0p5to1mos
- Map_RecoveryPotential_0to0p5mos
- Map_RecoveryPotential_1p5to2mos
- Map_RecoveryPotential_1to1p5mos
- Map_RecoveryPotential_2p5to3mos
- Map_RecoveryPotential_2to2p5mos
- Map_RecoveryPotential_3p5to4mos
- Map_RecoveryPotential_3to3p5mos
- Map_RecoveryPotential_4p5to5mos
- Map_RecoveryPotential_4to4p5mos
- Map_RecoveryPotential_5p5to6mos
- Map_RecoveryPotential_5to5p5mos
- Map_RecoveryPotential_6p5to7mos
- Map_RecoveryPotential_6to6p5mos
- Map_RecoveryPotential_7p5to8mos
- Map_RecoveryPotential_7to7p5mos
- Map_RecoveryPotential_8p5to9mos
- Map_RecoveryPotential_8to8p5mos
- Map_RecoveryPotential_9p5to10mos
- Map_RecoveryPotential_9to9p5mos
- Map_Structural_Complete_Destruction
- Map_Structural_Major_Damage
- Map_Structural_Minor_Damage
- Map_Structural_Moderate_Damage
- Inundation_BuildingMasked

Preview of the contents of the Masterfile.csv

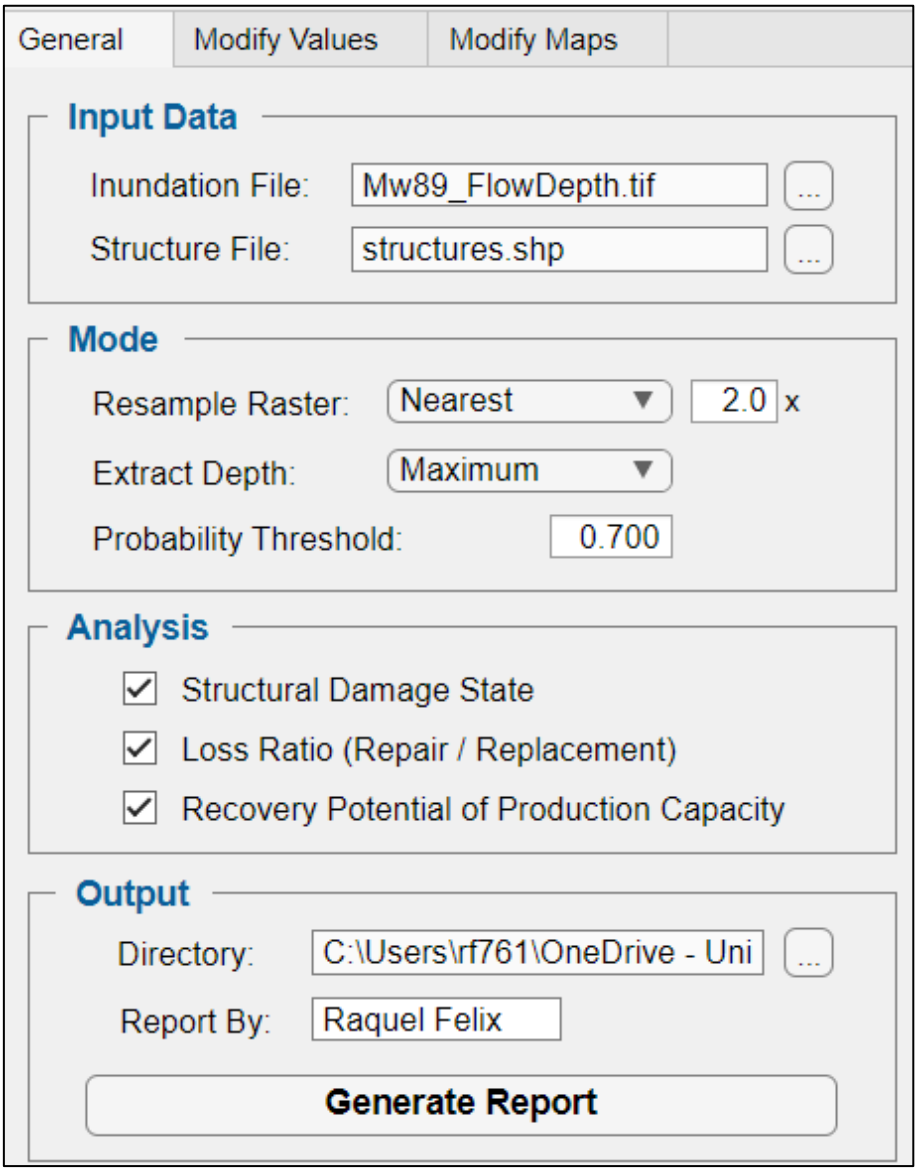
Full name of the file: *0_Polygon_Masterfile.csv*.

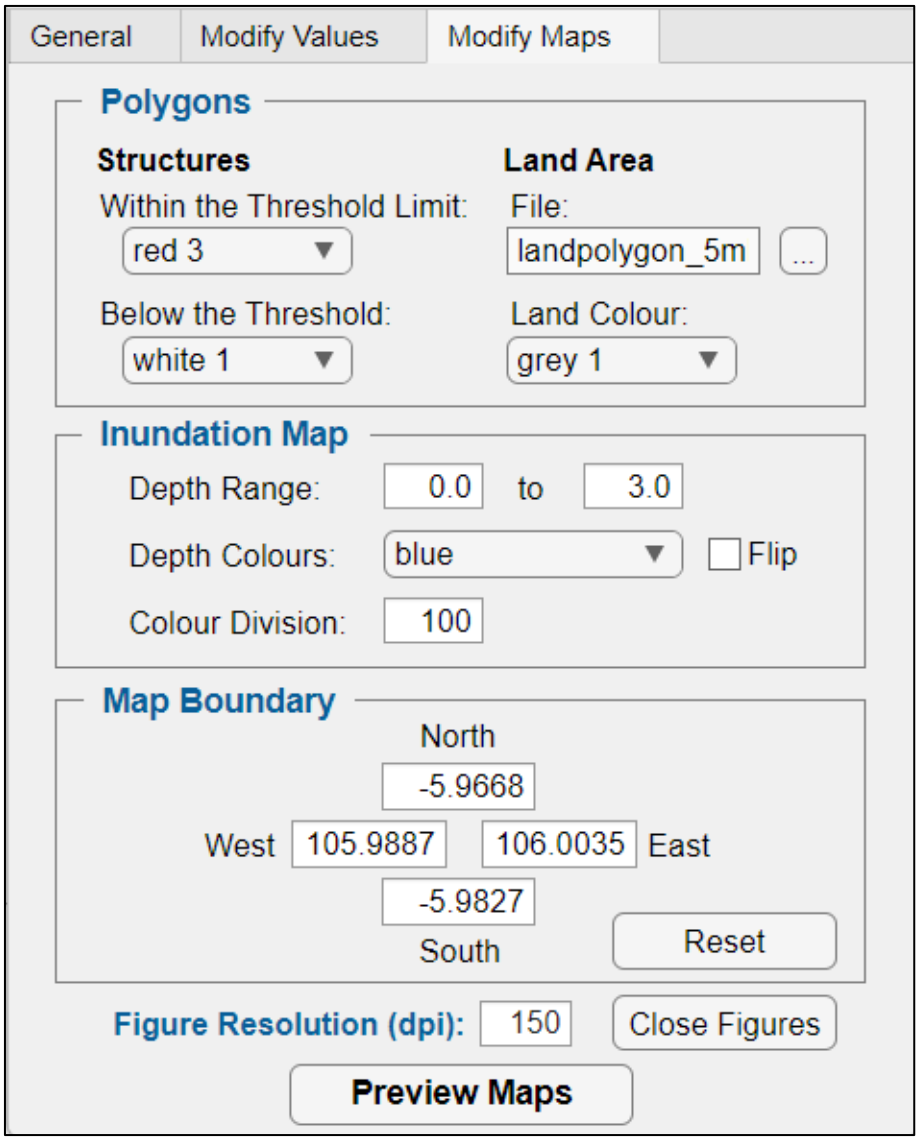
Each row (each structure) includes its inundation depth and the probability-of-exceedance values for the three risk metrics, calculated using the corresponding depth group.

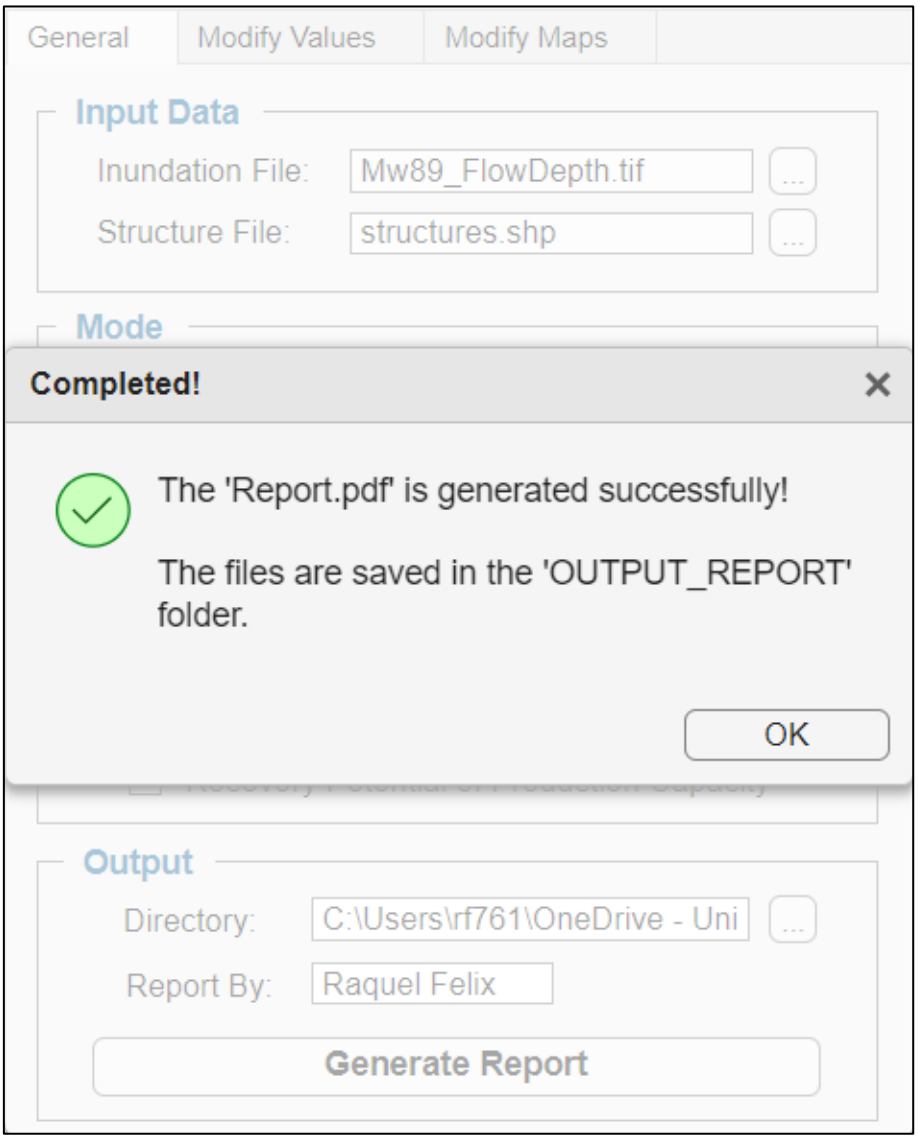
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	osm_id	name_3	type	ID	Inundation	DepthGrou	minor_Stru	moderate_	major_Stru	destroyed_	LossRatio	Recov	RecoveryP	Recove
2	5.13E+08	PT. Mitsubi	chemical	1	1.012255	1	0.8988	0.0488	0.0129	0.0042	0.4818	0	0.005	0.02
3	5.13E+08	PT. Mitsubi	chemical	2	1.289765	1.5	0.9731	0.171	0.0501	0.0171	0.5623	0	0.005	0.02
4	5.13E+08	PT. Mitsubi	chemical	3	1.286977	1.5	0.9731	0.171	0.0501	0.0171	0.5623	0	0.005	0.02
5	5.13E+08	PT. Mitsubi	chemical	4	1.273891	1.5	0.9731	0.171	0.0501	0.0171	0.5623	0	0.005	0.02
6	5.13E+08	Terminal B	petrochem	5	1.753518	2	0.9979	0.1276	0.0381	0.01	0.6098	0	0.005	0.02
7	5.13E+08	Terminal B	petrochem	6	1.616618	2	0.9979	0.1276	0.0381	0.01	0.6098	0	0.005	0.02
8	5.13E+08	Terminal B	petrochem	7	1.648692	2	0.9979	0.1276	0.0381	0.01	0.6098	0	0.005	0.02
9	5.13E+08	Terminal B	petrochem	8	2.248222	2.5	0.9992	0.266	0.0895	0.0244	0.7091	0	0.005	0.02
10	5.13E+08	Terminal B	petrochem	9	2.196474	2.5	0.9992	0.266	0.0895	0.0244	0.7091	0	0.005	0.02
11	5.13E+08	Terminal B	petrochem	10	2.382382	2.5	0.9992	0.266	0.0895	0.0244	0.7091	0	0.005	0.02
12	5.13E+08	Terminal B	petrochem	11	2.703122	3	0.9996	0.4328	0.1718	0.0499	0.8007	0	0.005	0.02
13	5.13E+08	Terminal B	petrochem	12	2.60987	3	0.9996	0.4328	0.1718	0.0499	0.8007	0	0.005	0.02
14	5.13E+08	Terminal B	petrochem	13	2.562668	3	0.9996	0.4328	0.1718	0.0499	0.8007	0	0.005	0.02

Preview of the generated Report.pdf

These are the input values used to generate the report that are shown on the subsequent pages.




General
Modify Values
Modify Maps
Polygons
Structures
Within the Threshold Limit:
red 3
Below the Threshold:
white 1
Land Area
File:
landpolygon_5m
...
Land Colour:
grey 1
Inundation Map
Depth Range:
0.0
to
3.0
Depth Colours:
blue
Flip
Colour Division:
100
Map Boundary
North
-5.9668
West
105.9887
106.0035
East
-5.9827
South
Reset
Figure Resolution (dpi):
150
Close Figures
Preview Maps


General
Modify Values
Modify Maps
Input Data
Inundation File:
Mw89_FlowDepth.tif
Structure File:
structures.shp
Mode
Completed!
The 'Report.pdf' is generated successfully!
The files are saved in the 'OUTPUT_REPORT' folder.
OK
Output
Directory:
C:\Users\rf761\OneDrive - Uni
Report By:
Raquel Felix
Generate Report

Tsunami Risk Assessment Report

generated using RAFT-Ports by Felix, et al. (2026)

Contents:

Inundated Structures
Structural Damage States
Physical Loss Ratio (Repair / Replacement)
Recovery of Production Capacity

Input Files:

Inundation: Mw89_FlowDepth.tif
File Resolution: 4.4870e-05, 4.4870e-05 degree
Structures: structures.shp

Raster Resampling Method: Nearest
Resampling Factor: 2.00
Updated Resolution: 2.243516e-05, 2.243486e-05 degree
Depth extracted within the building mask: Maximum
Probability Threshold: 0.70
Only structures meeting or exceeding this threshold are counted in the results

Report created by: Raquel Felix
Date created: 14-Aug-2026

RAFT-Ports: Risk Assessment for Tsunamis in Ports. RAFT-Ports provides scenario-based tsunami impact assessments to support risk-informed decision-making. Results are subject to uncertainties in input data, model assumptions, and the fragility and vulnerability functions used in the analysis. Users should independently review and validate all inputs and results prior to application.

Overview of the Results

Based on the analysis of 430 structures

Inundated Structures

total	chemical	petrochemical
418	334	84

Structural Damage States

minor	moderate	major	destroyed
416	0	0	0

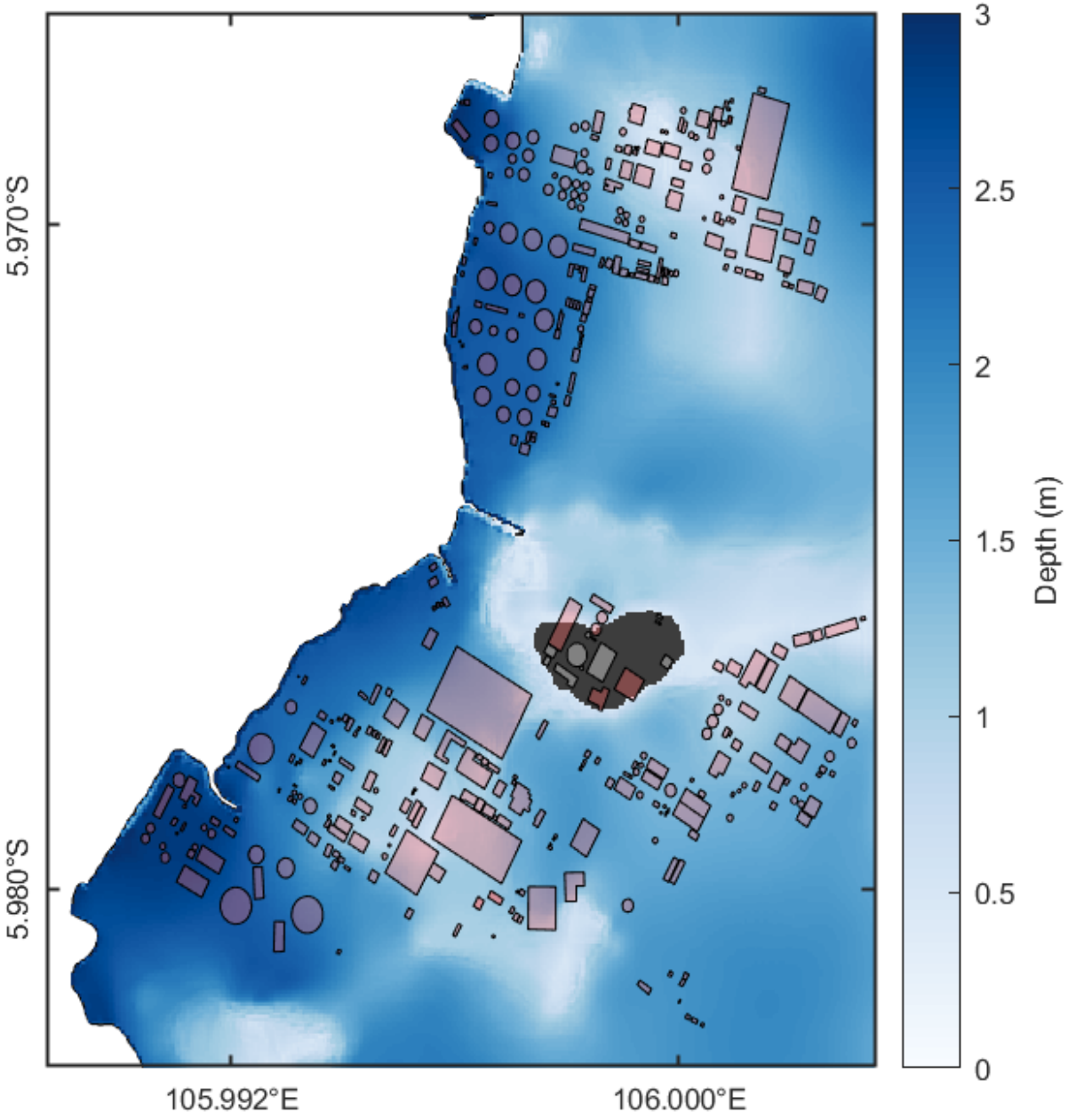
Physical Loss Ratio (Repair / Replacement)

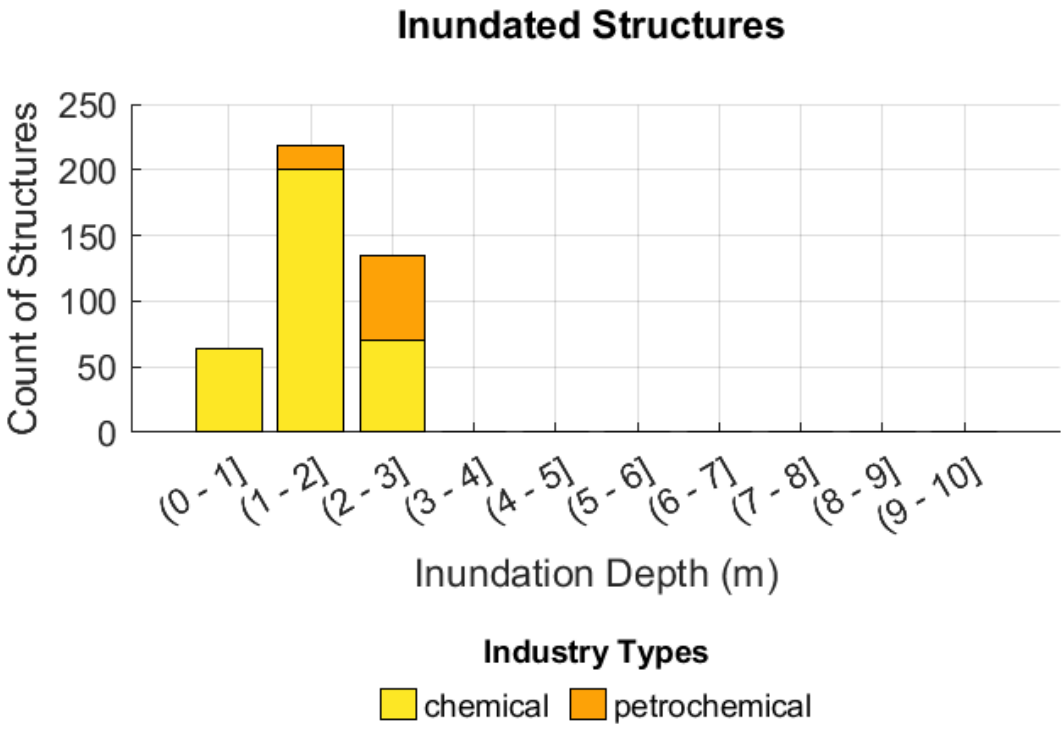
total	chemical	petrochemical
135	70	65

Potential Recovery Period of Production Capacity

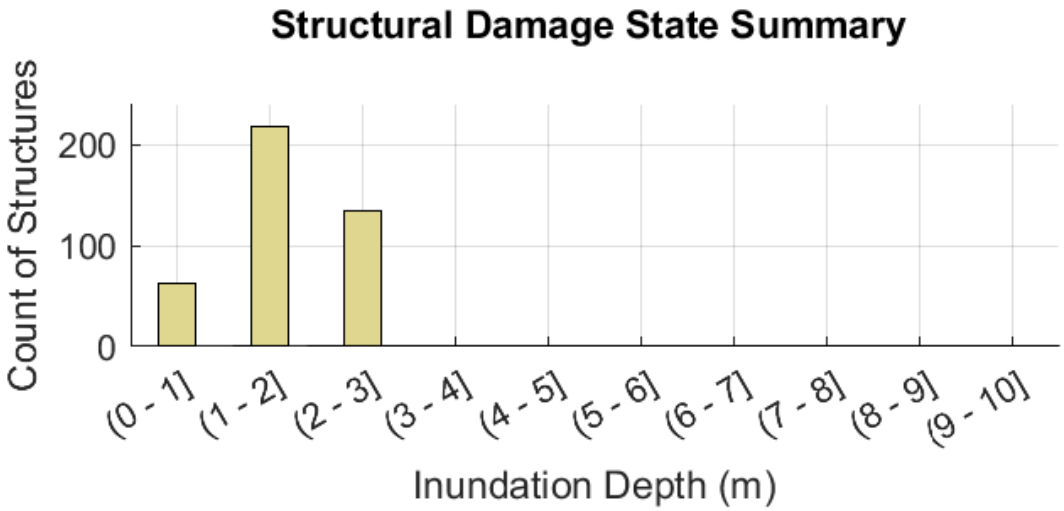
Potential Recovery	Month	Structure Count
Start of Recovery	(2.0 - 2.5]	12
Full Recovery	(5.0 - 5.5]	430

Structures Affected by Inundation





depth	total	chemical	petrochemical
(0 - 1]	64	64	0
(1 - 2]	219	200	19
(2 - 3]	135	70	65
(3 - 4]	0	0	0
(4 - 5]	0	0	0
(5 - 6]	0	0	0
(6 - 7]	0	0	0
(7 - 8]	0	0	0
(8 - 9]	0	0	0
(9 - 10]	0	0	0



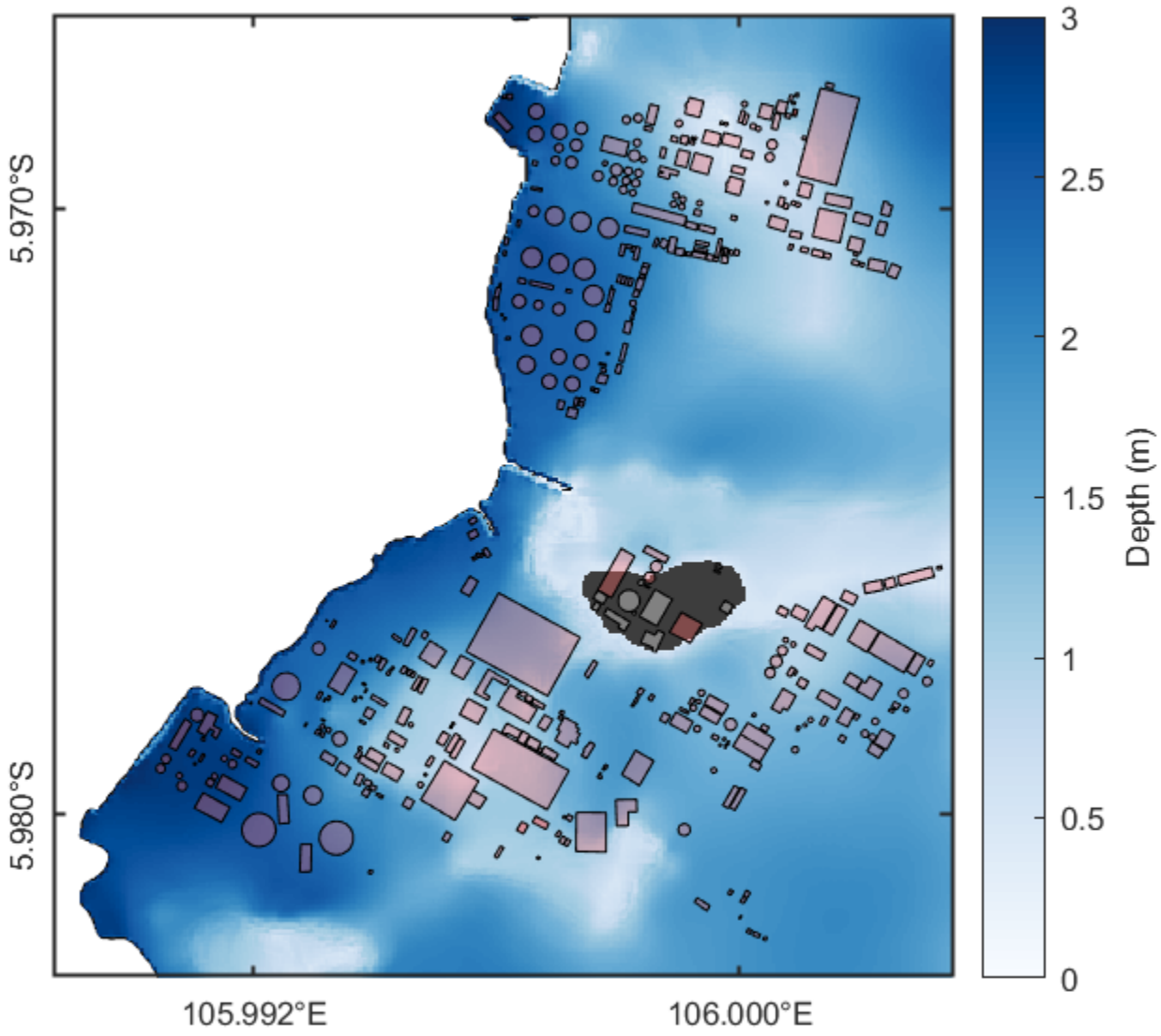
Count of Structures with Probability of Exceedance ≥ 0.70

Damage State

minor moderate major destroyed

depth	minor	moderate	major	destroyed
(0 - 1]	62	0	0	0
(1 - 2]	219	0	0	0
(2 - 3]	135	0	0	0
(3 - 4]	0	0	0	0
(4 - 5]	0	0	0	0
(5 - 6]	0	0	0	0
(6 - 7]	0	0	0	0
(7 - 8]	0	0	0	0
(8 - 9]	0	0	0	0
(9 - 10]	0	0	0	0

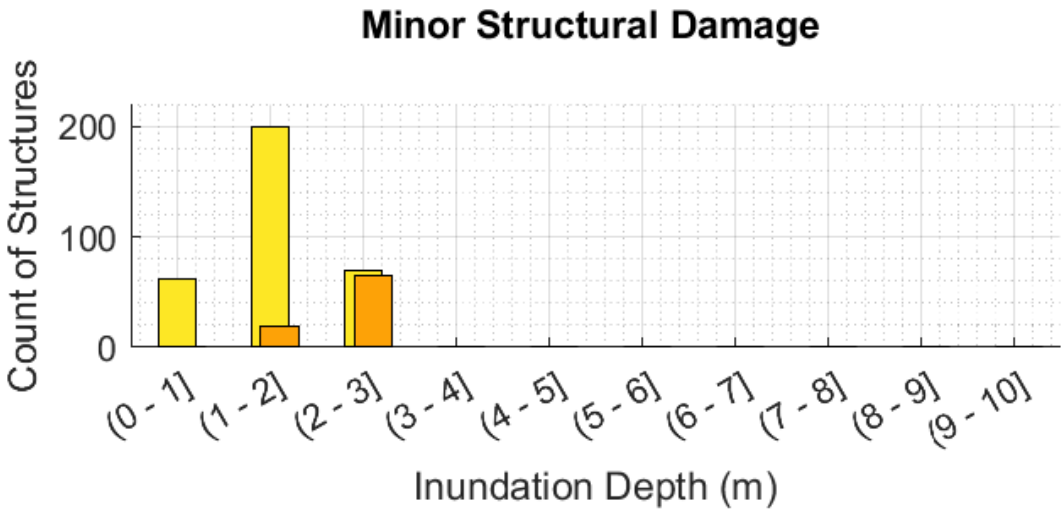
Minor Structural Damage



Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 416 structures have a probability of exceedance ≥ 70% for the selected damage state.

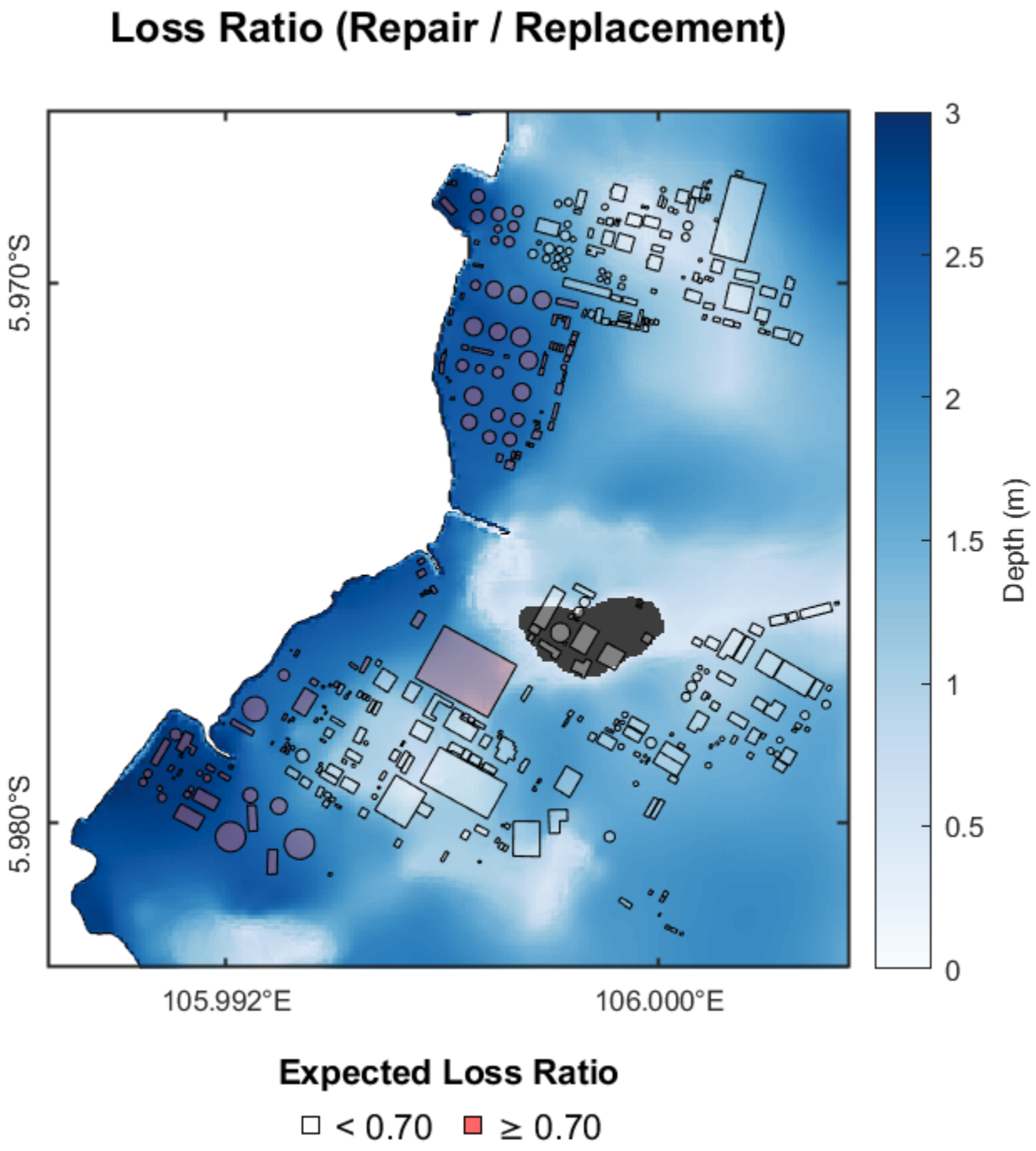


Count of Structures with Probability of Exceedance ≥ 0.70

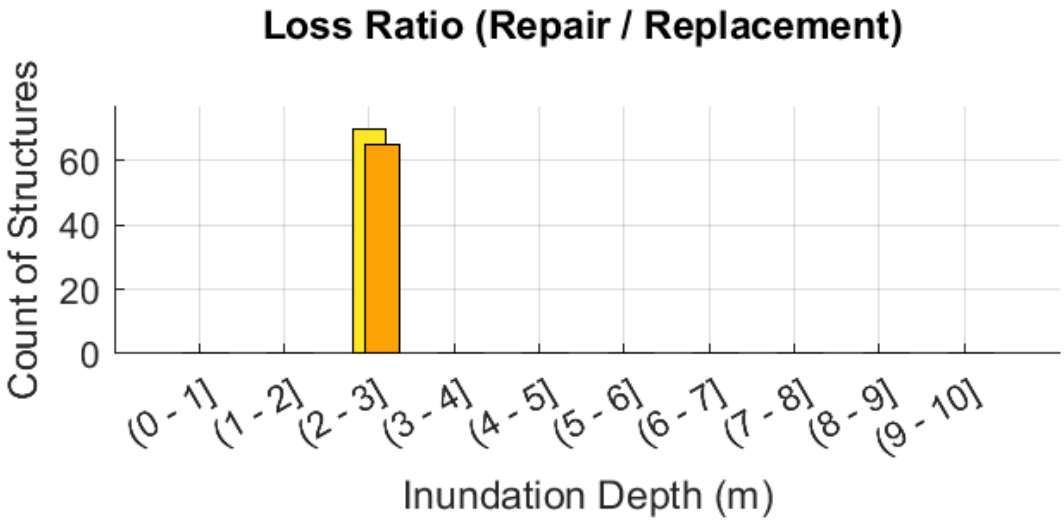
Industry Type

Chemical Petrochemical

depth	Total	chemical	petrochemical
(0 - 1]	62	62	0
(1 - 2]	219	200	19
(2 - 3]	135	70	65
(3 - 4]	0	0	0
(4 - 5]	0	0	0
(5 - 6]	0	0	0
(6 - 7]	0	0	0
(7 - 8]	0	0	0
(8 - 9]	0	0	0
(9 - 10]	0	0	0



A total of 135 structures have an expected loss ratio of ≥ 70%.



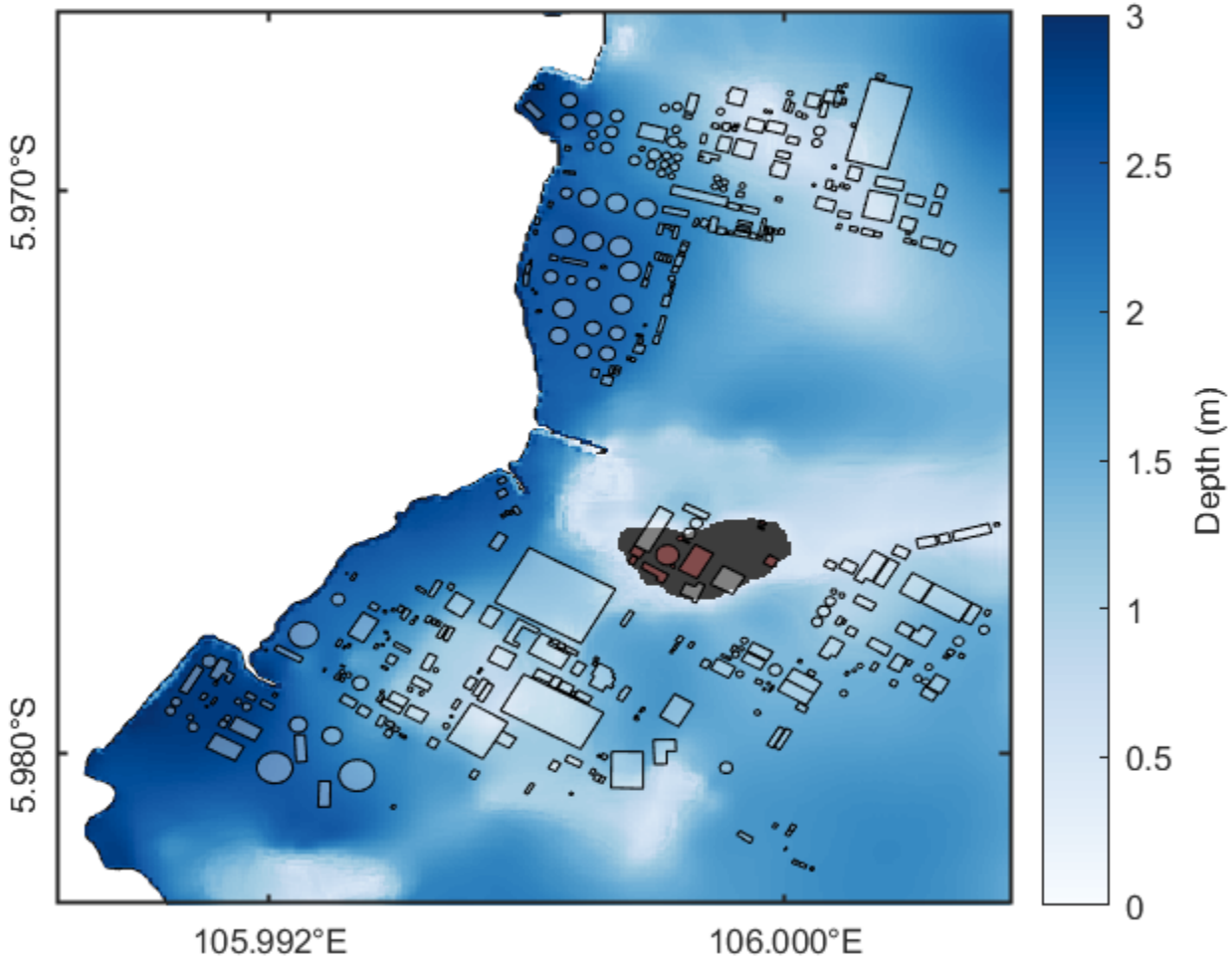
Count of structures with an expected loss ratio ≥ 0.70

Industry Type

chemical petrochemical

Depth	Total	chemical	petrochemical
(0 - 1]	0	0	0
(1 - 2]	0	0	0
(2 - 3]	135	70	65
(3 - 4]	0	0	0
(4 - 5]	0	0	0
(5 - 6]	0	0	0
(6 - 7]	0	0	0
(7 - 8]	0	0	0
(8 - 9]	0	0	0
(9 - 10]	0	0	0

Structures with a probability of recovery exceeding the threshold within 2.0 to ≤2.5 months

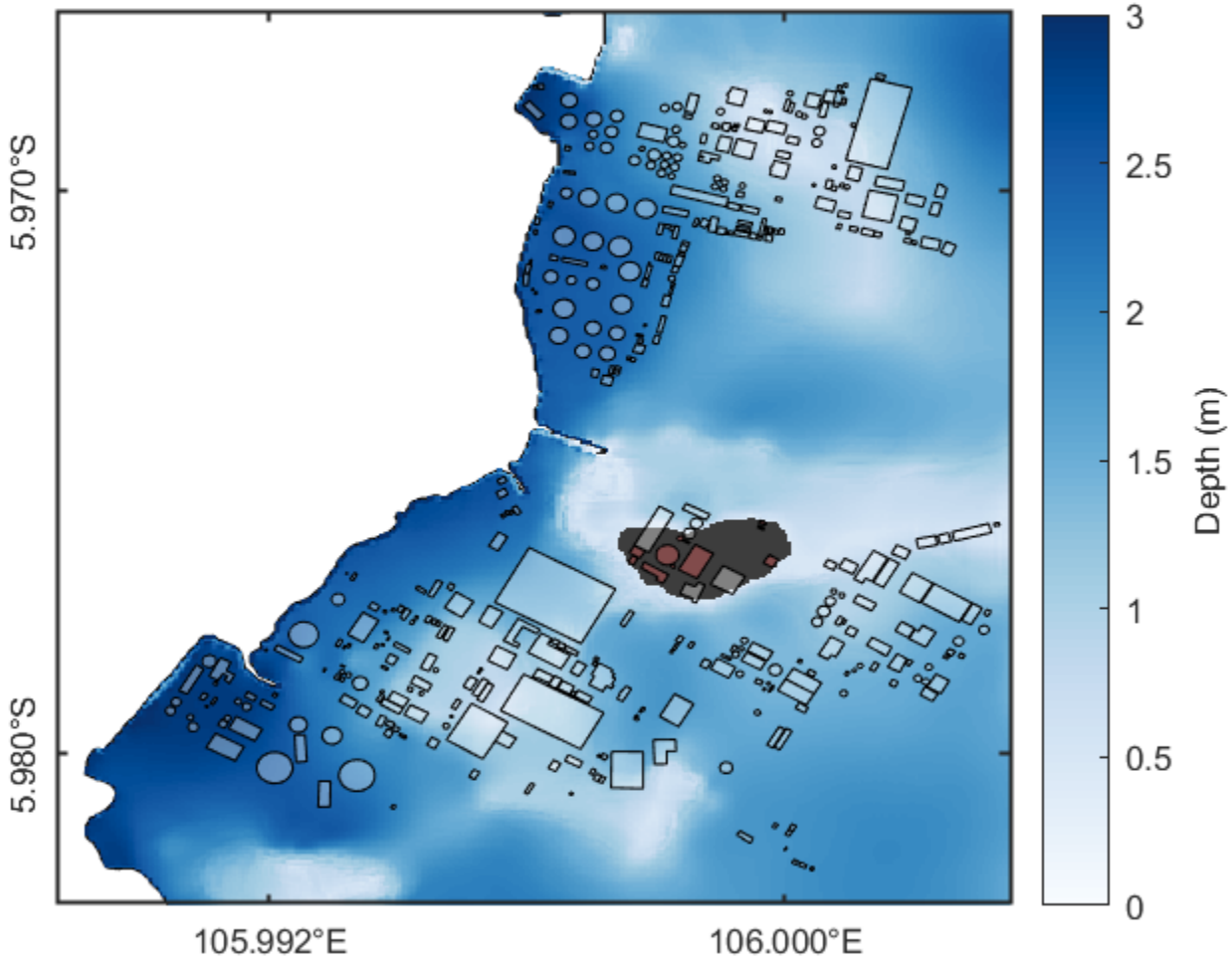


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 2.5 to ≤3.0 months

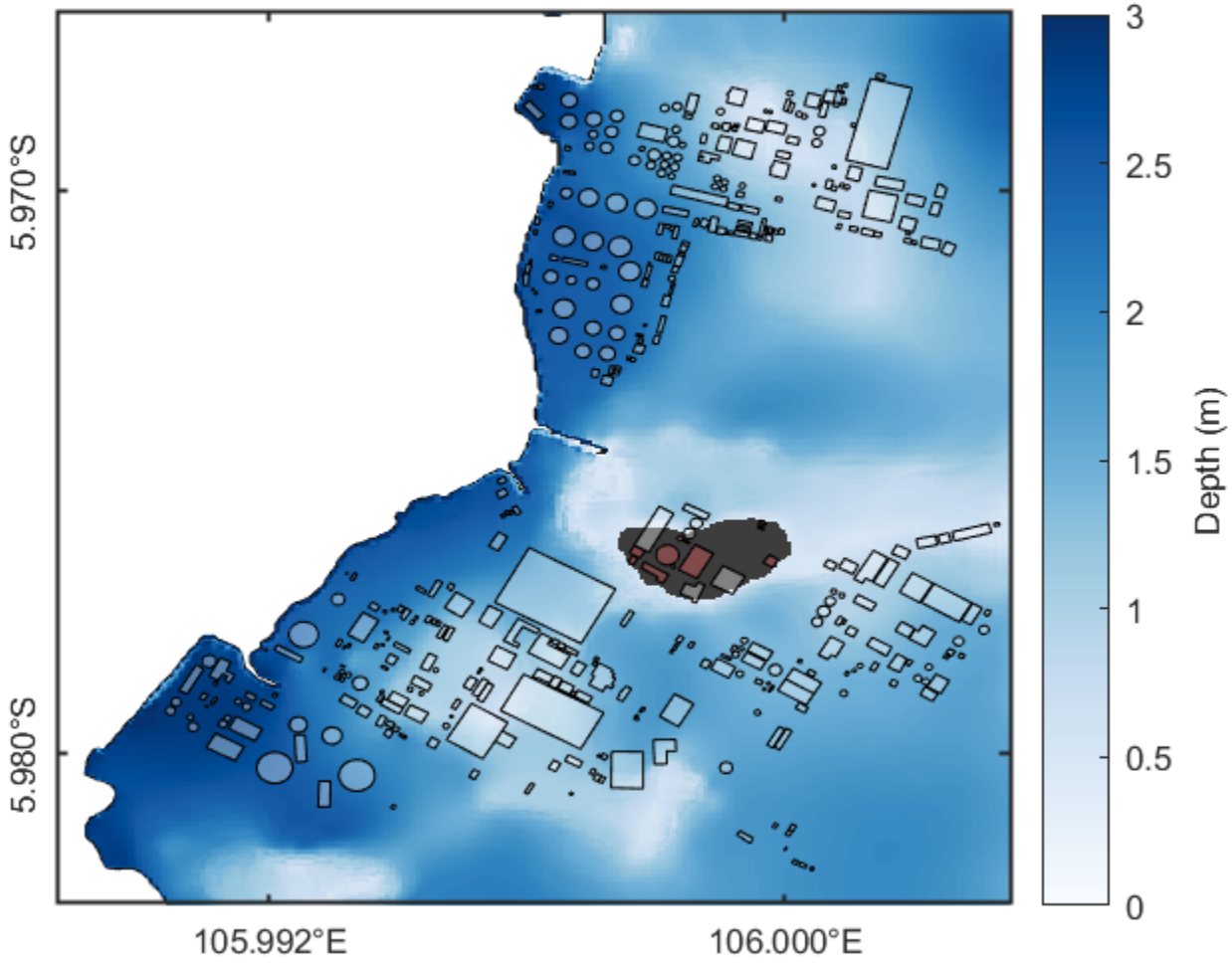


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 3.0 to ≤3.5 months

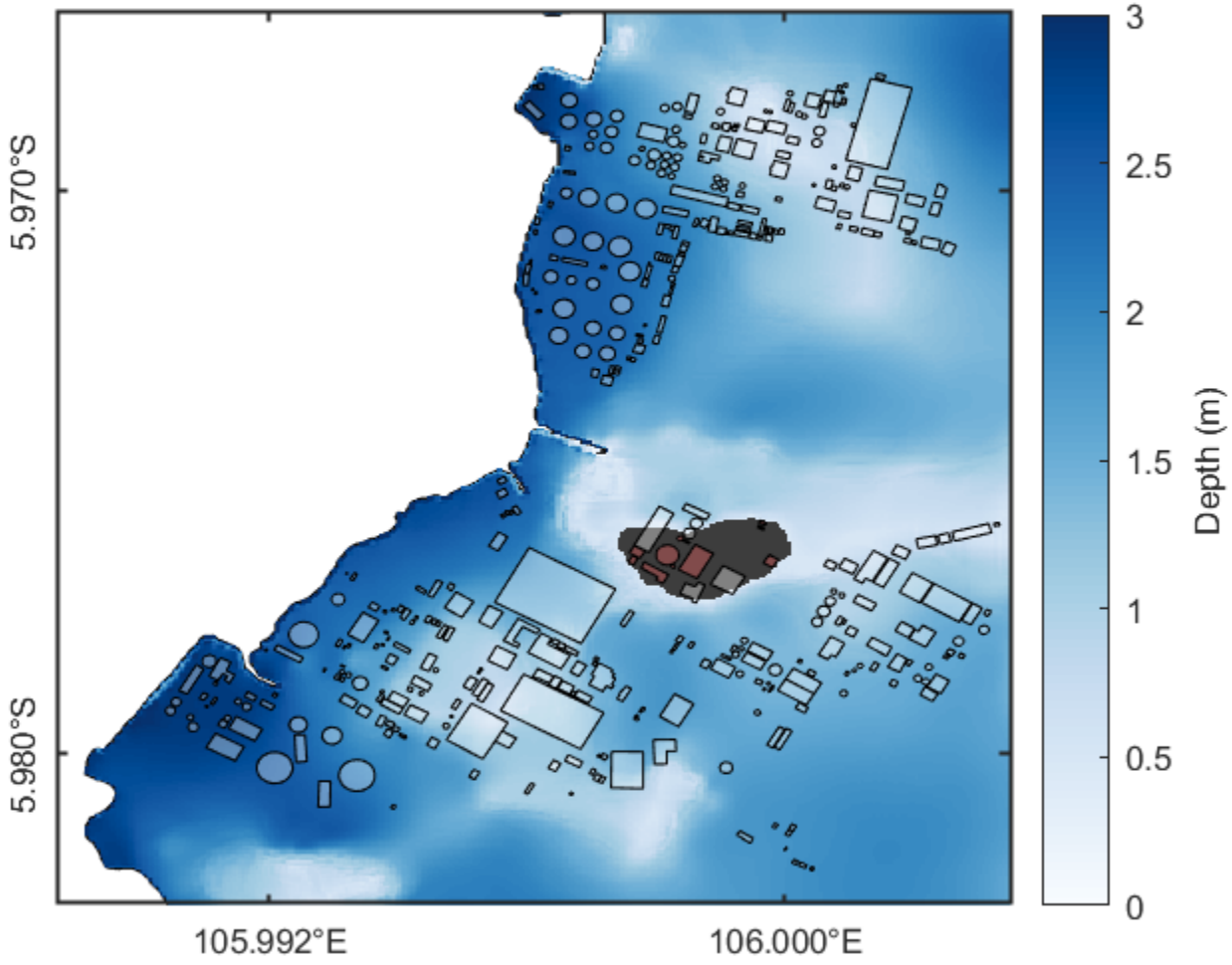


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 3.5 to ≤4.0 months

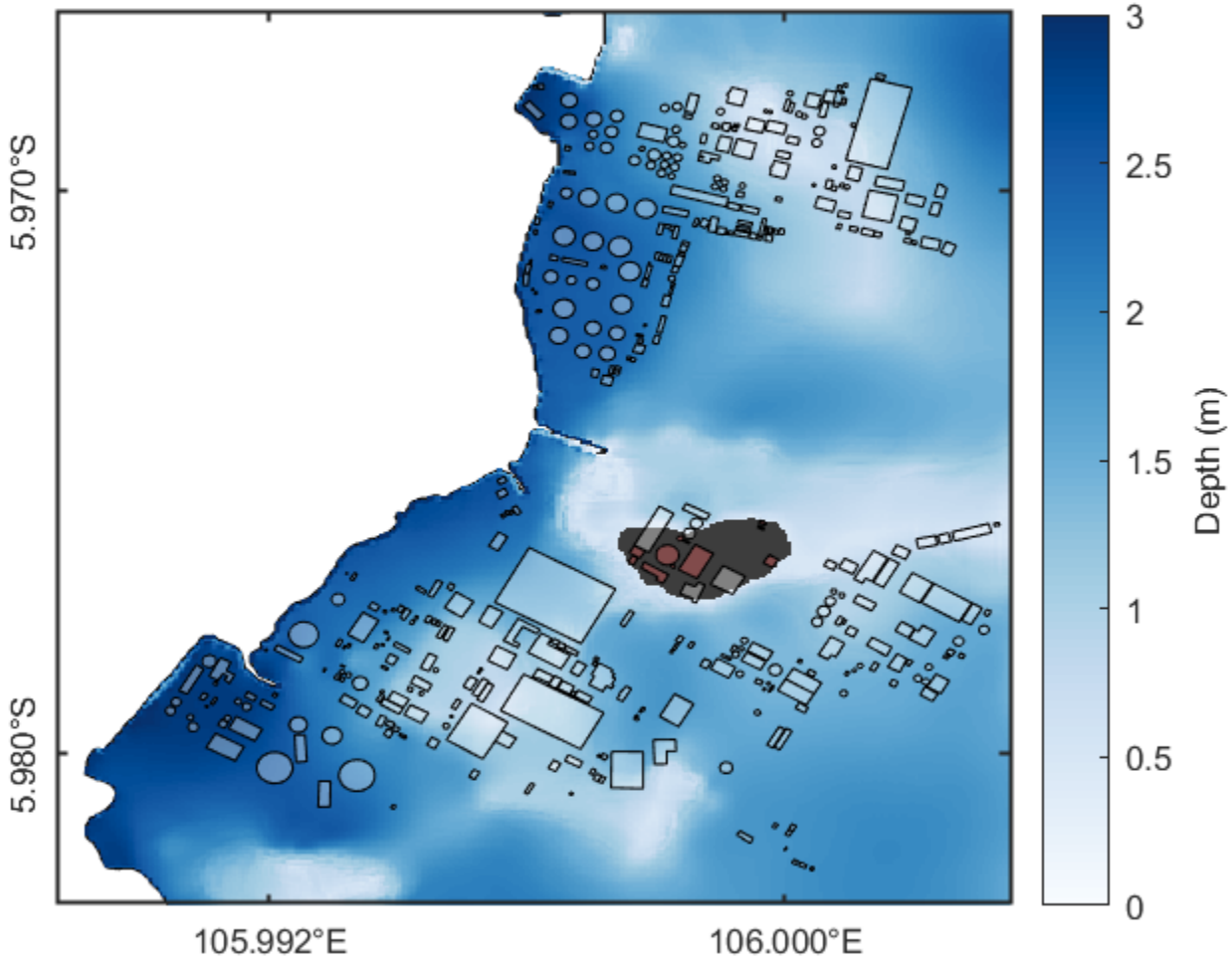


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 4.0 to ≤4.5 months

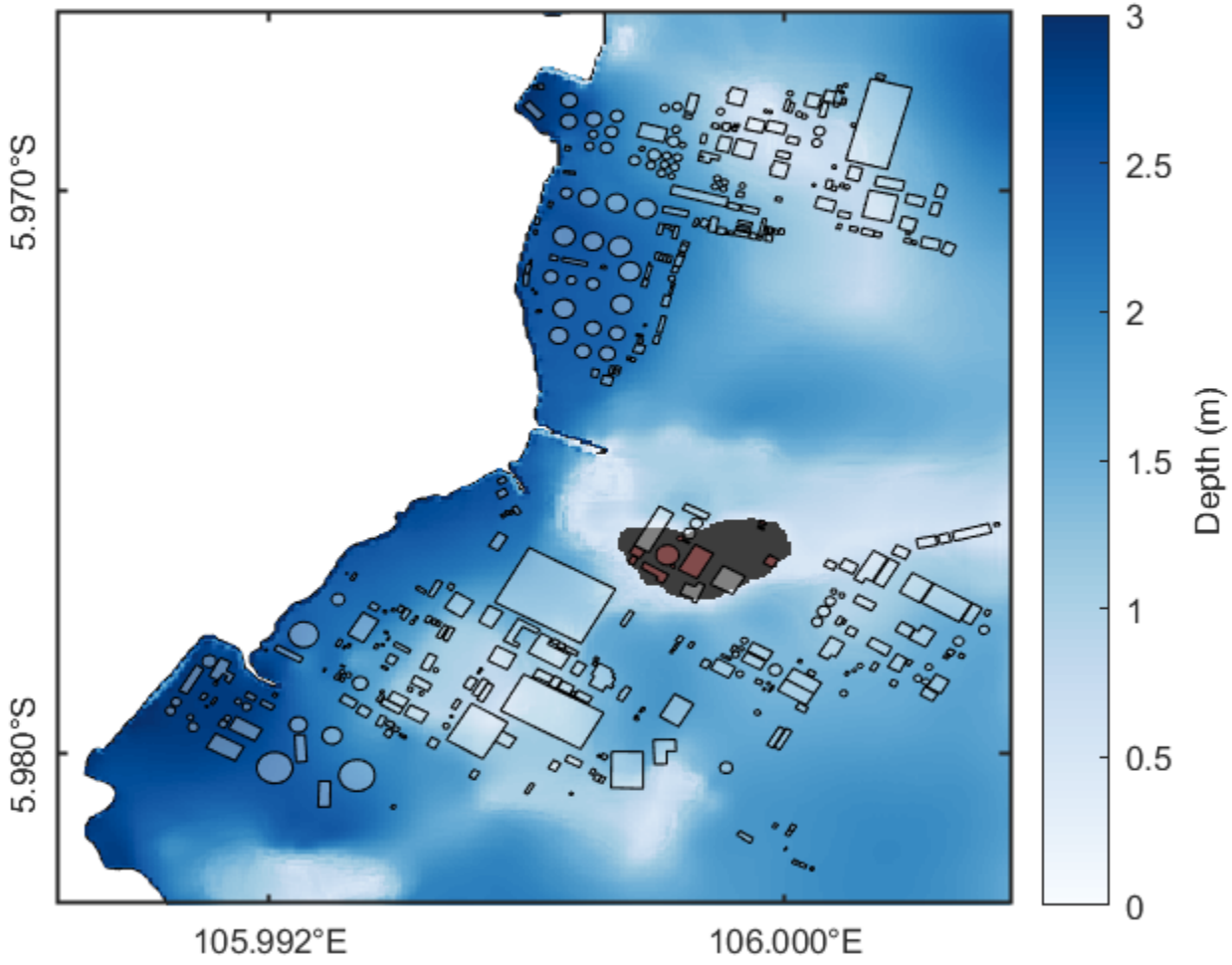


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 4.5 to ≤5.0 months

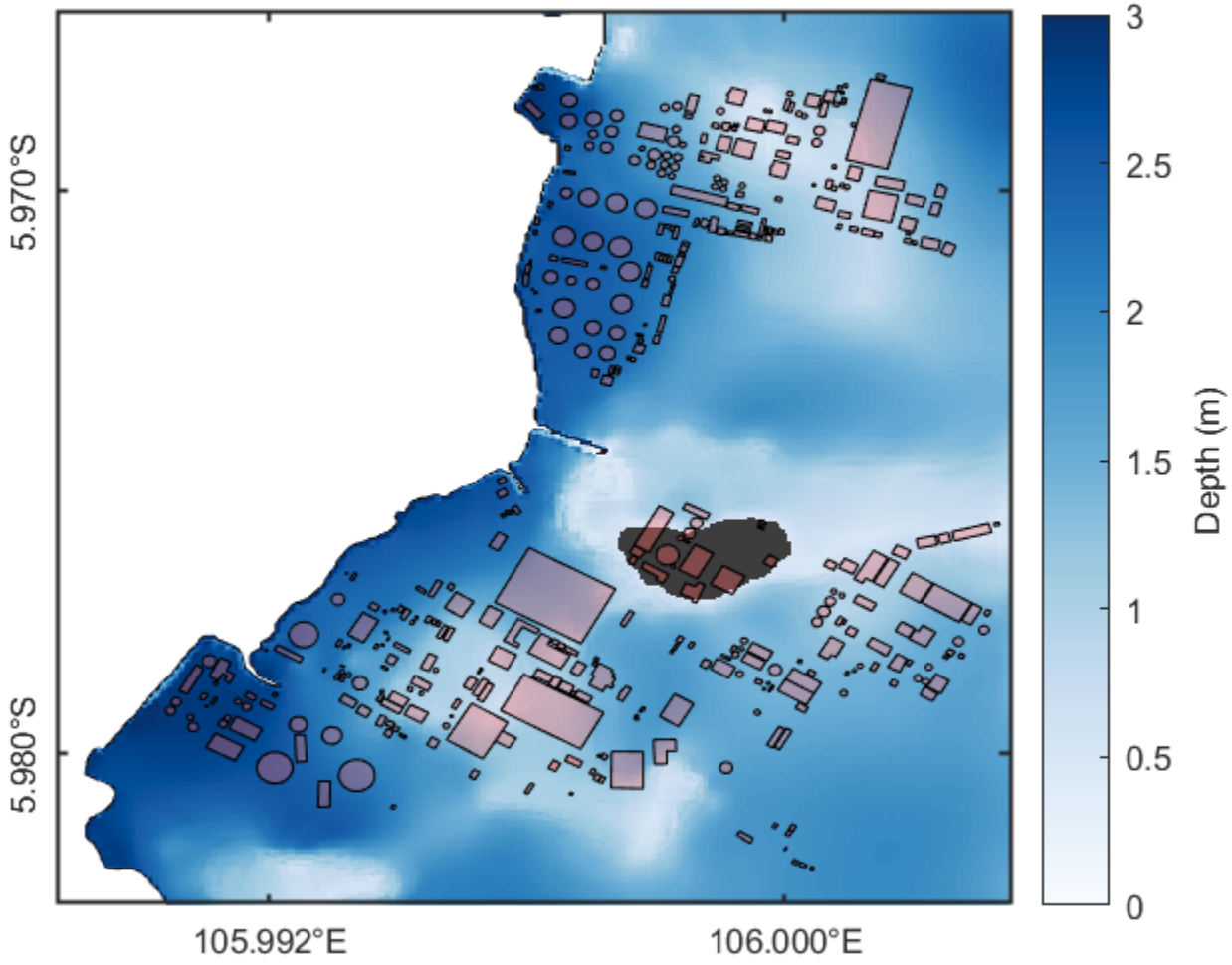


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 5.0 to ≤5.5 months

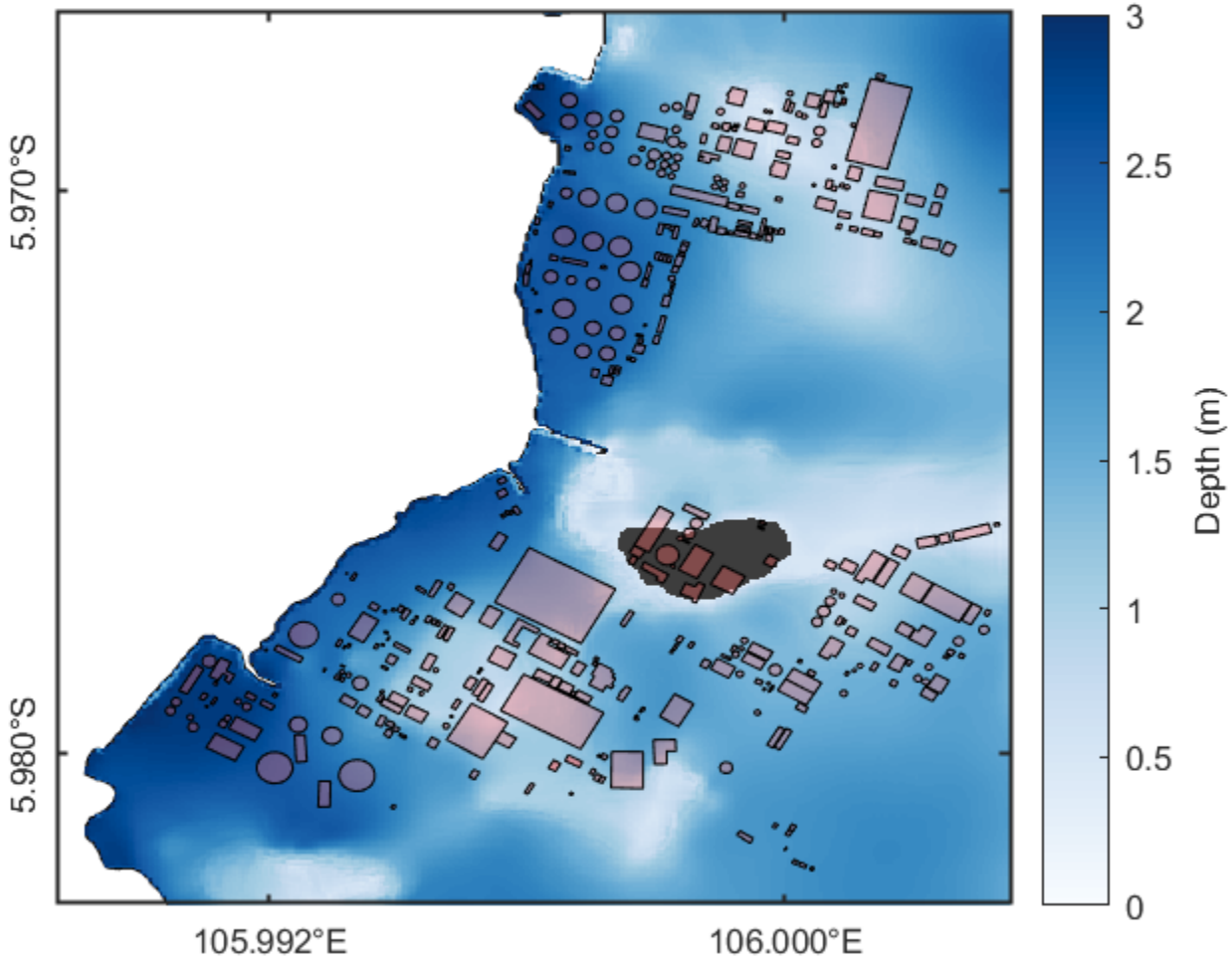


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 5.5 to ≤6.0 months

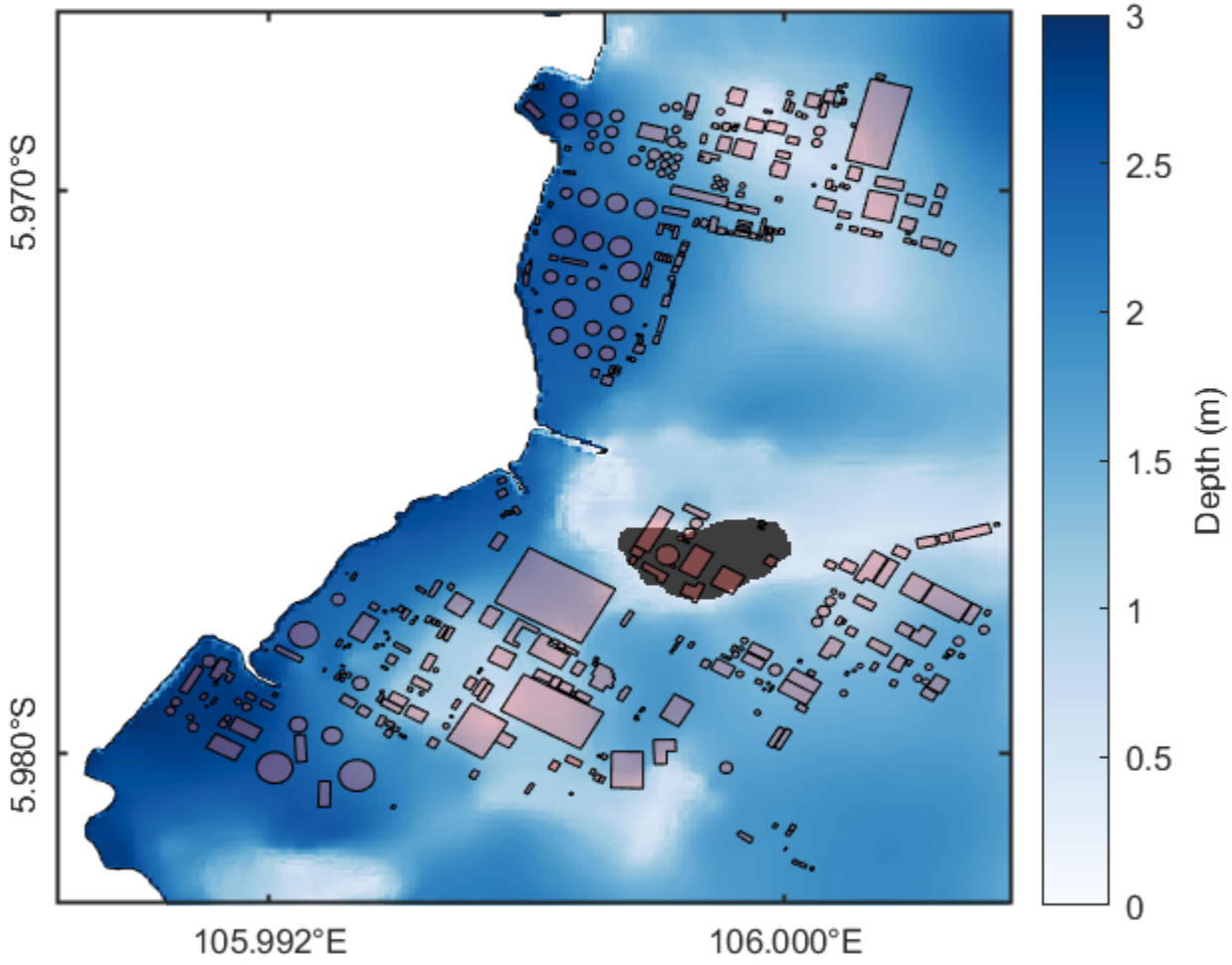


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 6.0 to ≤6.5 months

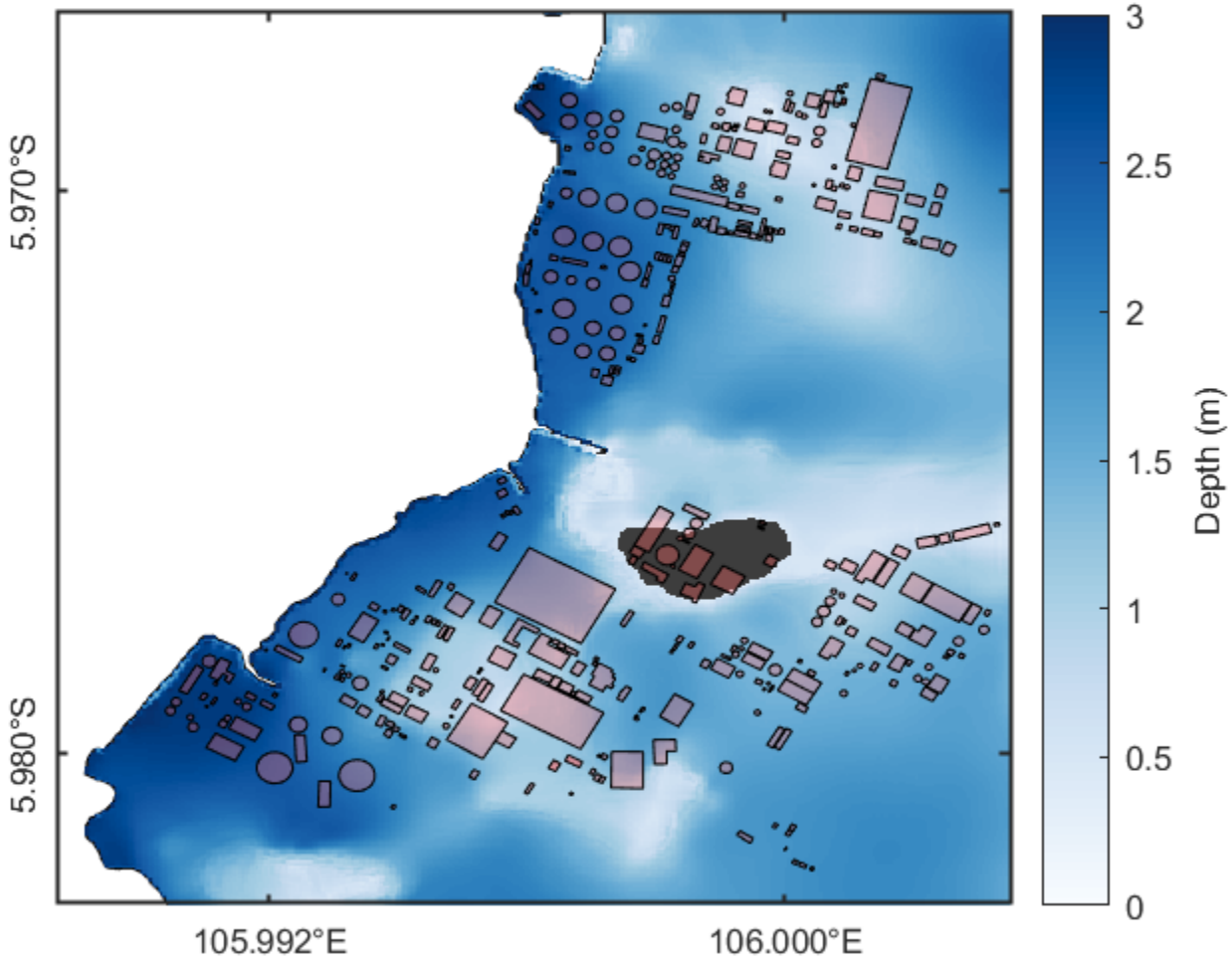


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 6.5 to ≤7.0 months

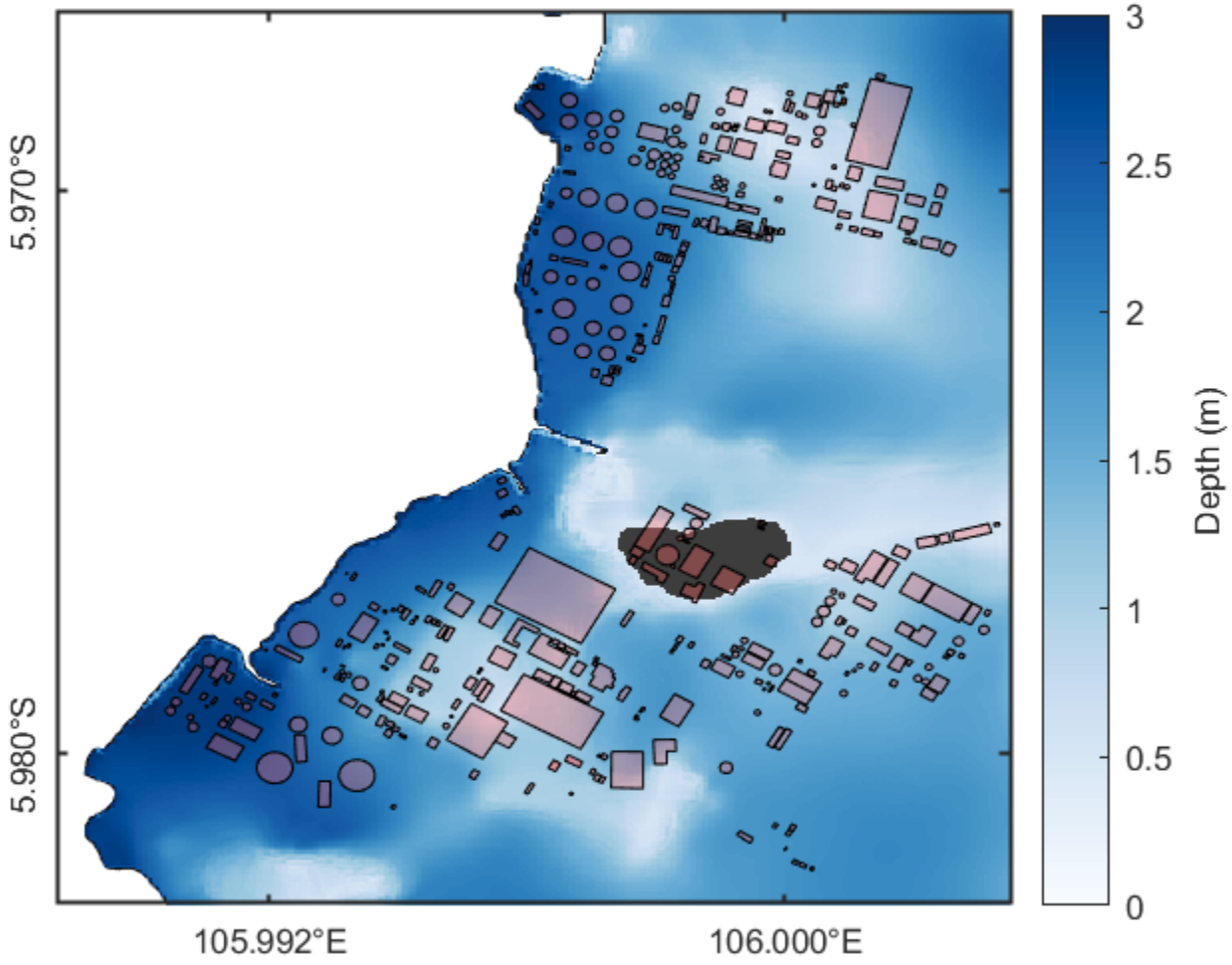


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 7.0 to ≤7.5 months

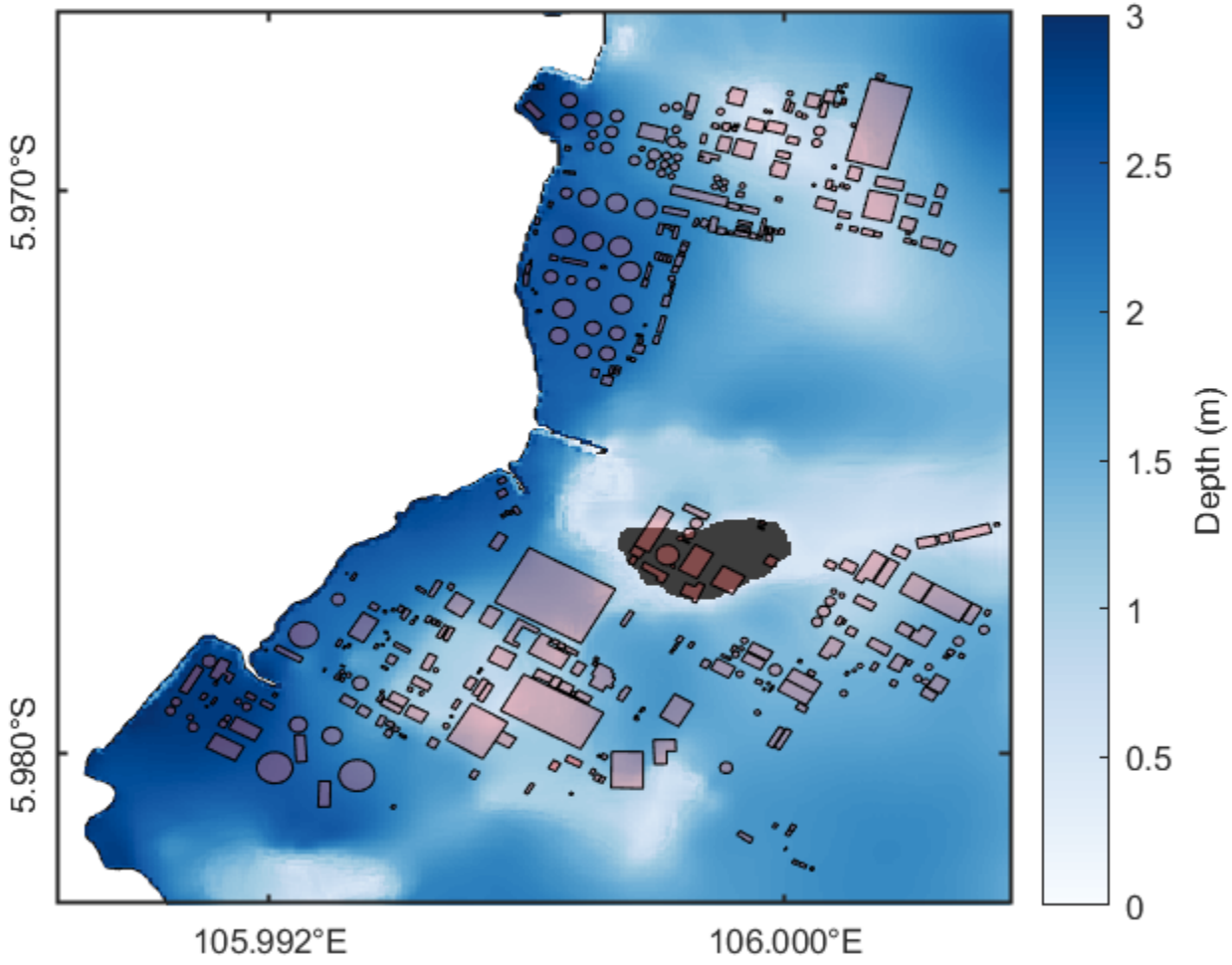


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 7.5 to ≤8.0 months

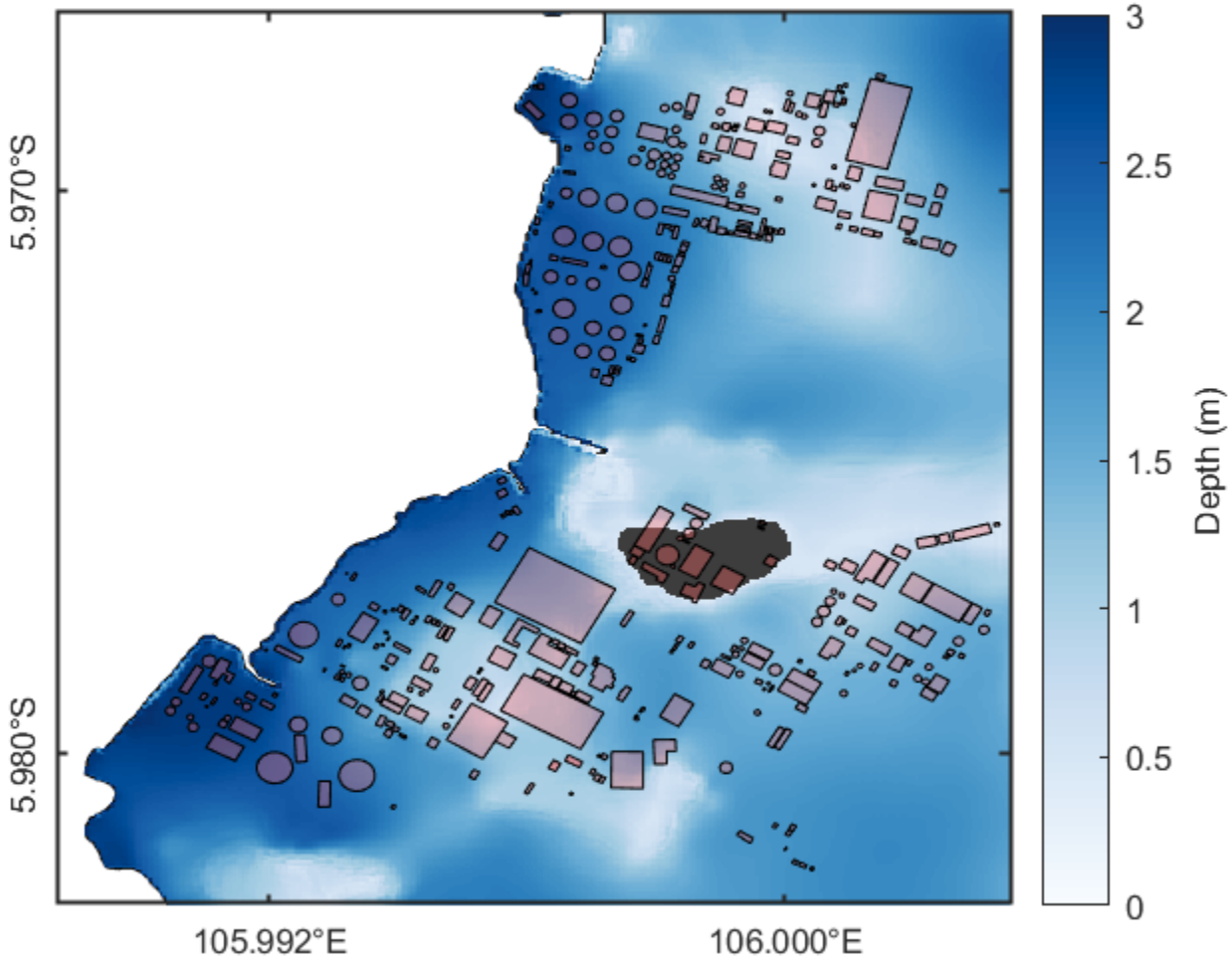


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 8.0 to ≤8.5 months

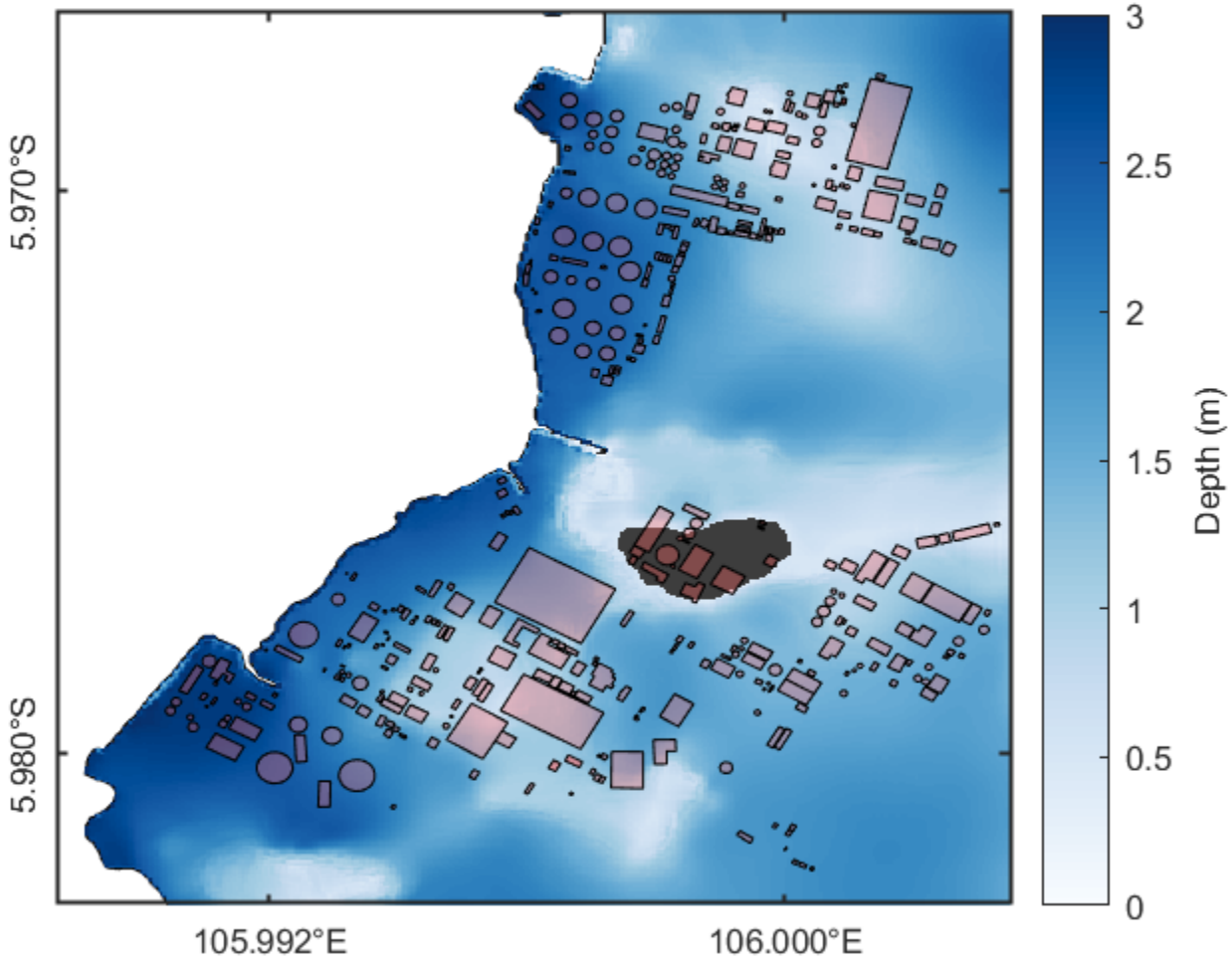


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 8.5 to ≤9.0 months

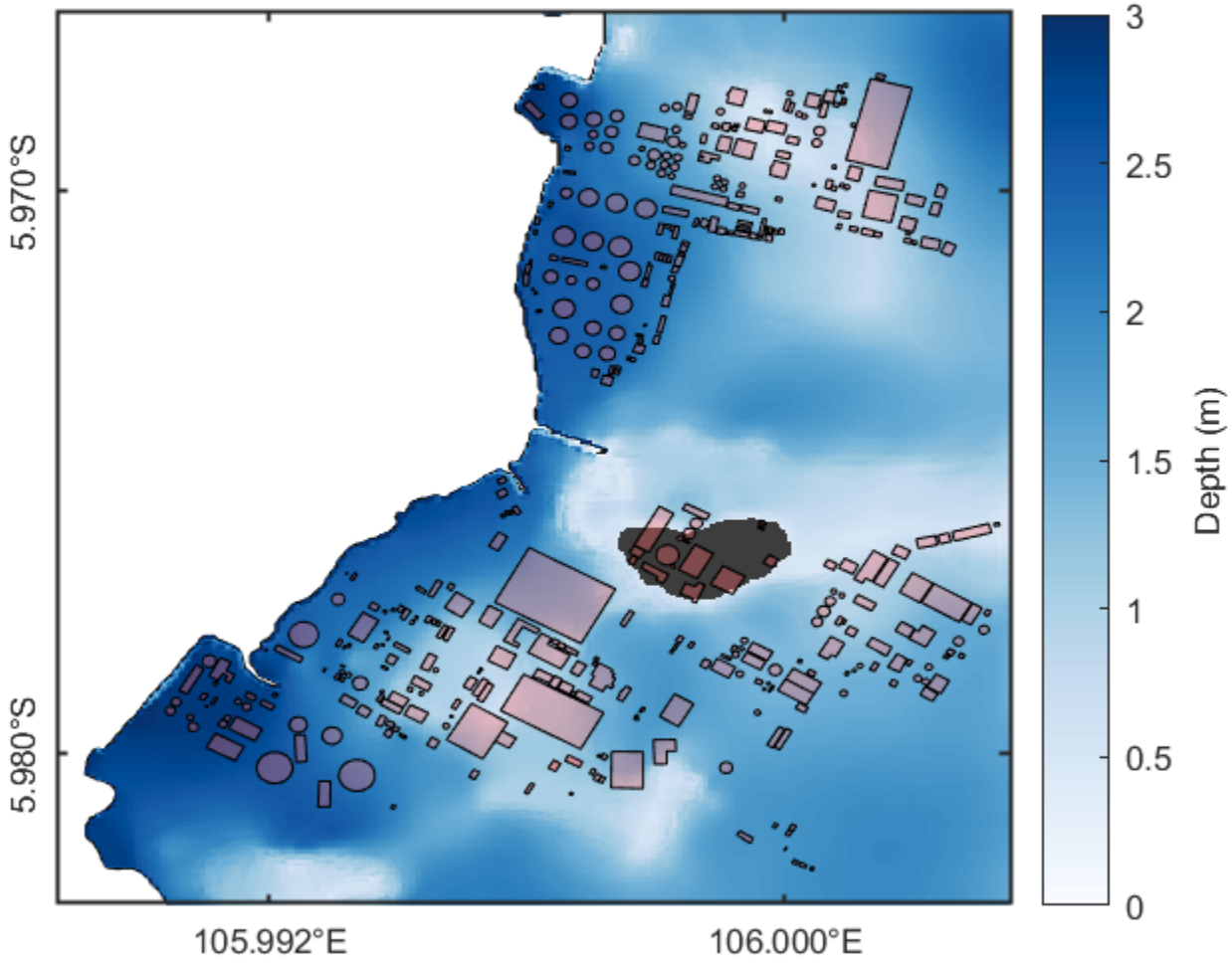


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 9.0 to ≤9.5 months

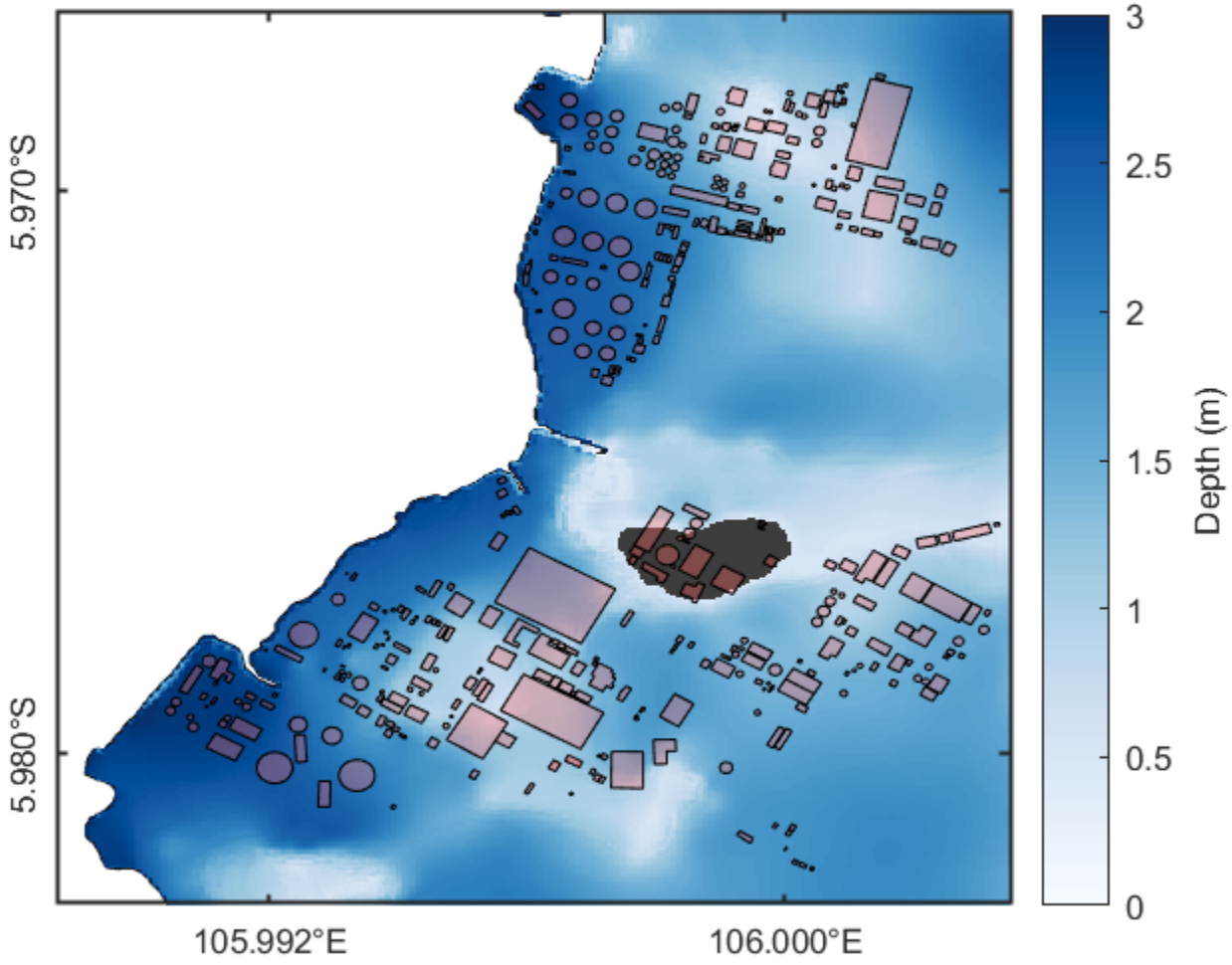


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

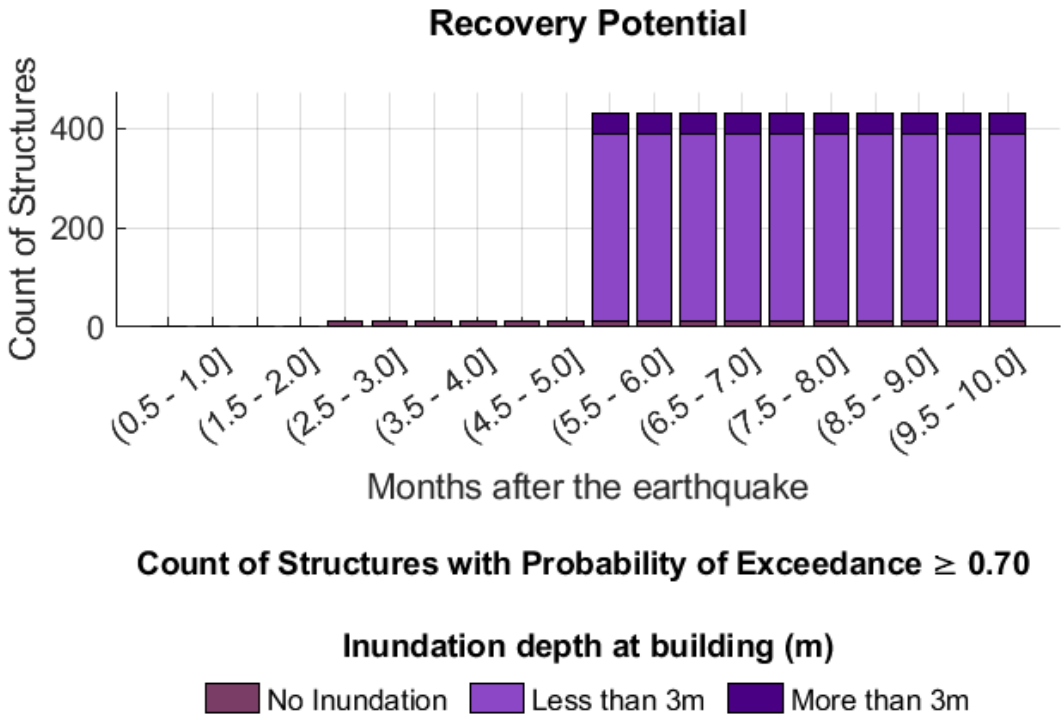
Structures with a probability of recovery exceeding the threshold within 9.5 to ≤10.0 months



Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.


Recovery Potential
Count of Structures
400
200
0
(0.5 - 1.0]
(1.5 - 2.0]
(2.5 - 3.0]
(3.5 - 4.0]
(4.5 - 5.0]
(5.5 - 6.0]
(6.5 - 7.0]
(7.5 - 8.0]
(8.5 - 9.0]
(9.5 - 10.0]
Months after the earthquake
Count of Structures with Probability of Exceedance ≥ 0.70
Inundation depth at building (m)
No Inundation
Less than 3m
More than 3m

Months	Total Count	At 0 m	At ≤3m	At >3m
(0.0 - 0.5]	0	0	0	0
(0.5 - 1.0]	0	0	0	0
(1.0 - 1.5]	0	0	0	0
(1.5 - 2.0]	0	0	0	0
(2.0 - 2.5]	12	12	0	0
(2.5 - 3.0]	12	12	0	0
(3.0 - 3.5]	12	12	0	0
(3.5 - 4.0]	12	12	0	0
(4.0 - 4.5]	12	12	0	0
(4.5 - 5.0]	12	12	0	0
(5.0 - 5.5]	430	12	376	42
(5.5 - 6.0]	430	12	376	42
(6.0 - 6.5]	430	12	376	42
(6.5 - 7.0]	430	12	376	42
(7.0 - 7.5]	430	12	376	42
(7.5 - 8.0]	430	12	376	42
(8.0 - 8.5]	430	12	376	42
(8.5 - 9.0]	430	12	376	42
(9.0 - 9.5]	430	12	376	42
(9.5 - 10.0]	430	12	376	42